Design and Analysis of Sample Survey

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1 Introduction

Throughout reference shall be made on a population which is defined as well defined collection of elements say $X_1, X_2, \ldots, X_N$, where N is the population size. In most cases the population in reference will be finite but for purpose of theoretical advancement.
We will also consider conceptual infinite population. In defining population, it must always be in terms of four quantities namely; the content, the unit, the extent and time.
Our interest will be describing certain attributes or characteristics or features or properties of a given target population and we use x_i to denote an arbitrary numerical attribute which can be determined for a particular population elements such as age, gender, status e.t.c.

The problem sample survey will seek to address is that of estimating the value of a population characteristics, that is either population mean, population proportion, population total from a subset(sample).

1.1 Elements of a Survey Design

- Specify the target population.
- Specify the sampling unit.
- Specify the precision and objectives of survey and declare other variables of interest (auxiliary information, include explanatory variables, stratification variables).
- Instruments we use.
- Develop or decide on sampling design/sampling scheme.
- Determine the sample size allocation.
- Explain the data processing procedures.
- Give an outline of the anticipated final report.

1.2 Types of Sampling

1. Haphazard sampling: - No scheme has been used at all it is neither probability nor non-probability sampling.
2. Purposive or judgemental sampling or non probability sampling.

3. Probability, random sampling - statistical theory is used and the kind of inferences made are based on statistical procedures. The simplest type of SRS(simple random sampling)

We shall also make use of common terms in statistics like; statistics, estimator, point estimation and interval estimation, ratio and regression estimation.

Sample survey: - It is a discipline concerned with identifying good estimators (estimation that are in some sense close to the population value that they intend to estimate) and determining sample designs which ensures that estimates are as required. Design and analysis of sample survey is about knowing those estimators and design procedures that are good.

1.3 Properties of Estimators

1.3.1 Precision:

- how much variation there is in the estimation from sample to sample.

1.3.2 Trueness:

- on average how close is the estimate to the population characteristics being estimated.

1.3.3 Accuracy:

- combination of precision and trueness. precision of estimates will be measured by variance e.g if the estimator is X then;

$$\text{var}(x) = \sigma_x^2 = E(X - \mu_x)^2$$

where $\mu_x = E(X)$

Trueness of an estimation will be measured by the bias, which is defined as a difference between the expectation of estimate and population parameter of which is an estimate.

$$\text{Bias}(x) = E(X) - \mu_x$$

If $E(X) - \mu_x = 0$ then it is unbiased estimator.

Accuracy will be measured using mean squared error.

$$\text{MSE}(x) = \sigma_x^2 + (\text{Bias}(x))^2$$

Generally estimator which have low variance (high precision) and low bias are preferred.

Remark: There are various methods for determining estimator: MME, OLS(ordinary least squares), MLE(maximum likelihood estimation), Bayesian estimation, Rao-Blackwell method, Minimum Chi-Square method and Minimum Distant method.

There exists several methods of comparing estimates, these include use of; bias, variance, MSE(mean squared error), consistency, sufficiency and location of scale invariance.

2 Simple Random Sampling (SRS)

This is a technique of selecting a sample from a target population in such a way that any unit in the population has an equal and independent chance of being selected into the sample. Therefore for a population of size N , the probability of picking a unit at first draw is $\frac{1}{N}$ and the second draw is $\frac{1}{N-1}$, third draw is $\frac{1}{N-2}$ hence for r^{th} draw is $\frac{1}{N-(r-1)}$.

In performing SRS one will do it with replacement or without replacement and in an event that it is with replacement then the above probabilities will be $\frac{1}{N}$ throughout.

2.1 Properties of Simple Random Sampling Without Replacement (SRSWOR)

The probability of a specified unit being chosen in the first draw is equal to the probability of choosing at any other draw and can be simply $\frac{1}{N}$.

If we consider the rth draw the probability of a specified unit being drawn at the rth draw is actually the probability of the event that is drawn from $(r-1)^{th}$ draws multiplied by the probability of the event that it is drawn at the rth draw.

Now the probability of not drawing an element:

$$\begin{aligned}
&\text{At the first draw is ..... } 1-\frac{1}{N}=\frac{N-1}{N}\\
&\text{At the second draw is ... } 1-\frac{1}{N-1}=\frac{N-2}{N-1}\\
&\text{At the third draw is..... } 1-\frac{1}{N-2}=\frac{N-3}{N-2}\\
&\vdots\\
&\text{At } (r-1)^{th} \text{ draw is..... } 1-\frac{1}{N-r+2}=\frac{N-r+1}{N-r+2}
\end{aligned}$$

and the probability of selecting at r^{th} draw is $\frac{1}{N-r+1}$ so accordingly the required probability will be,

$$\frac{N-1}{N}\cdot\frac{N-2}{N-1}\cdot\frac{N-3}{N-2}\cdots\frac{N-r+1}{N-r+2}\cdot\frac{1}{N-r+1}=\frac{1}{N}$$

The probability of a specified unit being chosen/selected into a sample of size n is $\frac{n}{N}$ (which is the sampling fraction f).

For a specified unit in a population N to be included in a sample of size n. It must have been drawn from the first draw or the nth draw, but the probability of a specified unit being drawn out of any draw is $\frac{1}{N}$ so the required probability is:

$$\Pr(\text{1st draw}) \text{ or } \Pr(\text{2nd draw}) \text{ or } \ldots \Pr(n^{th} \text{ draw})$$

$$\Rightarrow \frac{1}{N}+\frac{1}{N}+\frac{1}{N}+\cdots+\frac{1}{N}=\frac{n}{N}$$

The probability of selecting a sample of size n in a population of size N is $\frac{1}{{}^NC_n}$

From the definition of SRS the probability of selecting an element at first draw is $\frac{1}{N}$, second draw $\frac{1}{N-1}$ third draw $\frac{1}{N-2}$ and thus at the nth draw is equal to $\frac{1}{N-(n-1)}$. Thus in that specified order the probability of selecting a sample of size n from a population of size N is,

$$\frac{1}{N}\cdot\frac{1}{N-1}\cdot\frac{1}{N-2}\cdots\frac{1}{N-(n-1)}$$

But this order is immaterial and accordingly there exists $n!$ possible re-orderings which would lead to this required probability, so the exact probability that we wish to have is,

$$\frac{n!}{N(N-1)(N-2)\cdots(N-n+1)}=\frac{n!(N-n)!}{N!}$$

but it is known that ${}^NC_n=\frac{N!}{(N-n)!n!}$ so that $\frac{n!(N-n)!}{N!}=\frac{1}{{}^NC_n}$ which completes the proof.

In some context this property summarized is used as alternative definition of simple random sampling, particularly in a case where referring to SRSWOR. In this case the statement will be SRS is a sampling scheme in which a sample of size n is selected from a population of size N with the probability $\frac{1}{{}^NC_n}$.

2.2 Notations used in SRS

Accordingly;

Population parameters	Sample counterparts
Y or $T = \sum_{i=1}^{N} Y_i$ population total	y or $t = \sum_{i=1}^{n} y_i$ sample total
$\bar{Y} = \frac{1}{N}\sum_{i=1}^{N} Y_i$ population mean	$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$ sample mean
$S^2 = \frac{1}{N}\sum(Y_i - \bar{Y})^2$ population variance	$s^2 = \frac{1}{n-1}\sum(y_i - \bar{y})^2$ sample variance

Next let $a_i = \begin{cases} 1 & \text{if } i \in \text{sample} \\ 0 & \text{if } i \notin \text{sample} \end{cases}$

Using indicator variable a_i and the above notations it is easy to see that;

$$y = \sum_{i=1}^{n} y_i = \sum_{i=1}^{N} a_i Y_i$$

$$= \sum_{i=1}^{n} a_i Y_i + \sum_{i=n+1}^{N} a_i Y_i$$

$$= \sum_{i=1}^{n} y_i + 0$$

with the above information, we may study the next property of SRSWOR.

In SRSWOR $E(s^2) = S^2$ where $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2$.

Therefore

$$E(s^2) = \frac{1}{n}\sum_{i=1}^{N} Y_i^2\left(\frac{n}{N}\right) - \frac{1}{n(n-1)}\sum_{i\neq j}^{N} Y_i Y_j \frac{n(n-1)}{N(N-1)}$$

$$= \frac{1}{N}\sum_{i=1}^{N} Y_i^2 - \frac{1}{N(N-1)}\sum_{i\neq j}^{N} Y_i Y_j$$

$$= \frac{1}{N}\sum_{i=1}^{N} Y_i^2 - \frac{1}{N(N-1)}\left[(N\bar{Y})^2 - \sum_{i=1}^{N} Y_i^2\right]$$

$$= \left(\frac{1}{N} - \frac{1}{N(N-1)}\right)\sum_{i=1}^{N} Y_i^2 - \frac{N\bar{Y}^2}{(N-1)}$$

$$= \frac{1}{N-1}\sum_{i=1}^{N} Y_i^2 - \frac{N\bar{Y}^2}{(N-1)}$$

$$= \frac{1}{N-1}\left(\sum_{i=1}^{N} Y_i^2 - N\bar{Y}^2\right)$$

$$= S^2$$

In simple random sampling without replacement,

$$\text{var}(\bar{y}) = \frac{N-n}{N} \cdot \frac{S^2}{n}$$

From the definition of variance, $\text{var}(\bar{y}) = E(\bar{y}^2) - [E(\bar{y})]^2$. Now

$$\bar{y}^2 = \left(\frac{1}{n}\sum_{i=1}^{n} y_i\right)^2 = \left(\frac{1}{n}\sum_{i=1}^{N} a_i Y_i\right)^2$$

$$= \frac{1}{n^2}\sum_{i=1}^{N} a_i Y_i^2 + \frac{1}{n^2}\sum_{i\neq j}^{N} a_i a_j Y_i Y_j$$

Therefore:

$$= \left[\frac{1}{Nn} - \frac{n-1}{Nn(N-1)}\right]\sum_{i=1}^{N} Y_i^2 + \frac{(n-1)}{Nn(N-1)}(N\bar{Y})^2 - \bar{Y}^2$$

$$= \left[\frac{1}{Nn} - \frac{n-1}{Nn(N-1)}\right]\sum_{i=1}^{N} Y_i^2 - \left[\frac{Nn - N - Nn + n}{Nn(N-1)}\right]\bar{Y}^2$$

$$= \frac{N-n}{Nn(N-1)}\sum_{i=1}^{N} Y_i^2 - \frac{N-n}{Nn(N-1)}N\bar{Y}^2$$

$$= \frac{N-n}{Nn}\left[\frac{1}{N-1}\left\{\sum_{i=1}^{N} Y_i^2 - N\bar{Y}^2\right\}\right]$$

$$= \frac{N-n}{Nn}S^2$$

$$= \frac{N-n}{N} \cdot \frac{S^2}{n} \text{ or } (1-f)\frac{S^2}{n}$$

where f is the usual sampling fraction.

In $\text{var}(\bar{y})$ we have a population parameter S^2 which must be estimated using its sample counterpart s^2 as; $s^2 = \frac{1}{n-1}\sum(y_i - \bar{y})^2$. Because of this we conventionally write that:

$$\widehat{\text{var}}(\bar{y}) = \frac{N-n}{N} \cdot \frac{s^2}{n} \text{ or } (1-f)\frac{s^2}{n}$$

2.2.1 Exercise

Suppose that we are in SRSWR- (simple random sampling with replacement), what would be $\text{var}(\bar{y})$.

2.3 Remarks

1. The standard error may be computed using $\widehat{\text{var}}(\bar{y})$ as $\frac{s}{\sqrt{n}}\sqrt{1-f}$

2. If one wishes to compute confidence limits then $\bar{y}$ and $\hat{y}$ are assumed to be normally distributed about the population parameters.

3. With the normality assumption holding, the upper and the lower limits for the population mean are $\left(\bar{y} - \frac{ts}{\sqrt{n}}\sqrt{1-f}, \bar{y} + \frac{ts}{\sqrt{n}}\sqrt{1-f}\right)$ where t is the normal variate corresponding to the confidence probability desired. Similarly for total;

$$\left(N\bar{y} - \frac{tNs}{\sqrt{n}}\sqrt{1-f}, N\bar{y} + \frac{tNs}{\sqrt{n}}\sqrt{1-f}\right)$$

4. In many surveys estimates are meant for each of a number of classes into which population are subdivided. The term domain of study have been given to these sub populations in the simplest situation in which each unit falls into only one of the domains we then have SRSWOR, in an event that this doesn't hold then other sampling schemes have to be considered.

2.3.1 Illustative Example on SRS

In a private library, the books are kept on 130 shelves of similar size. The numbers of books on 15 shelves picked at random were found to be 28,23,25,33,31,18,22,29,30,22,26,20,21,28 and 25. Estimate the total number Y_T , of books in the library and calculate an approximate 95% confidence interval for Y_T.

Solution

$N = 130, n = 15$

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$$

$$= \frac{1}{15}(28 + 23 + 25 + 33 + 31 + 18 + 22 + 29 + 30 + 22 + 26 + 20 + 21 + 28 + 25)$$

$$= \frac{1}{15}(381)$$

$$= 25.4$$

$$Y_T = N\bar{y}$$

$$= 130 \times 25.4$$

$$= 3302$$

The 95% confidence interval:

$$\hat{Y} = N\bar{y} \pm \frac{tNs}{\sqrt{n}}\sqrt{1-f}$$

$$= Y_T \pm N z_{\alpha/2}\sqrt{\widehat{\text{var}}(\bar{y})}$$

but $S^2 = \frac{1}{N-1}\left(\sum_{i=1}^{N} Y_i^2 - N\bar{Y}^2\right)$ which is estimated by $s^2 = \frac{1}{n-1}\left(\sum_{i=1}^{n} y_i^2 - n\bar{y}^2\right)$

$$\sum_{i=1}^{15} y_i^2 = \left(28^2 + 23^2 + 25^2 + 33^2 + 31^2 + 18^2 + 22^2 + 29^2 + 30^2 + 22^2 + 26^2 + 20^2 + 21^2 + 28^2 + 25^2\right)$$

$$= 784+529+625+1089+961+324+484+841+900+484+676+400+441+784+625 = 9947$$

$$n\bar{y}^2 = 15 \times (25.4)^2 = 15 \times 645.16 = 9677.4$$

$$s^2 = \frac{1}{14}(9947 - 9677.4) = \frac{269.6}{14} = 19.257$$

$$\widehat{\text{var}}(\bar{y}) = \frac{130-15}{130} \cdot \frac{19.257}{15} = \frac{115}{130} \times 1.2838 = 0.8846 \times 1.2838 = 1.136$$

$$\sqrt{\widehat{\text{var}}(\bar{y})} = \sqrt{1.136} = 1.066$$

$$\sqrt{\widehat{\text{var}}(Y_T)} = 130 \times 1.066 = 138.58$$

95% CI: $3302 \pm 1.96 \times 138.58 = 3302 \pm 271.62$
Thus: $3030.38 \leq Y_T \leq 3573.62$

3 Stratified Random Sampling

The objective of any sampling method is usually to estimate the unknown population parameters with the highest precision i.e the variance of the estimators should be minimized. If the population is heterogeneous as will be in most situations then a sample taken via SRS might yield high levels of variability. As a result in a survey where precision is a main factor to be considered, then a strategy that addresses heterogeneity must be found. One way of achieving higher precision is to divide the population which is originally heterogeneous into sub population which are to a big extent homogeneous with respect to survey characteristics. These sub population are known as strata. Once the strata have been formed, we proceed to take a SRS from each stratum. From the sample taken from each of the stratum we compute sample means and sample variances thus we obtain estimators with better precision.

3.1 Notations Used:

Let population consist of N units. Divide the population into k strata in which the ith stratum consists of N_i units.

Let the strata be mutually disjoint so that $\sum_i N_i = N_1 + N_2 + \cdots + N_k = N$

After formation of the strata, let the sample sizes from each stratum be n_i such that $n_1 + n_2 + \cdots + n_k = n$
where n was the required size for the entire study.
Let Y_{ij} be the value of the characteristic for the jth unit in the ith stratum.

let $\bar{Y}_i$ be the ith stratum population mean this implies that $\bar{Y}_i = \frac{1}{N_i}\sum_{j=1}^{N_i} Y_{ij}$

let $\bar{y}_i$ be the sample mean of the ith stratum, $\bar{y}_i = \frac{1}{n_i}\sum_{j=1}^{n_i} y_{ij}$

let S_i^2 be the population variance in the ith stratum, $S_i^2 = \frac{1}{N_i - 1}\sum_{j=1}^{N_i}(Y_{ij} - \bar{Y}_i)^2$

Now the description from stratified random sampling, it is clear that.

$$E(\bar{y}_i) = \bar{Y}_i$$

$$E(s_i^2) = S_i^2$$

$$\text{var}(\bar{y}_i) = \frac{N_i - n_i}{N_i n_i} S_i^2$$

Proposition: $\bar{y}_w$ defined as

$$\bar{y}_w = \sum_{i=1}^{k} \frac{N_i}{N} \bar{y}_i = \sum_{i=1}^{k} w_i \bar{y}_i$$

can be used as an estimator of the overall population mean $\bar{Y}$

3.2 Theorem

$\bar{y}_w$ is an unbiased estimator of $\bar{Y}$ in the context of stratified random sampling

Proof: if $\bar{y}_w$ is unbiased for $\bar{Y}$ this implies $E(\bar{y}_w) = \bar{Y}$. Now

$$E(\bar{y}_w) = E\left(\sum_{i=1}^{k} w_i \bar{y}_i\right)$$

where $w_i = \frac{N_i}{N}$ which is the stratum weight.

$$\begin{aligned} E(\bar{y}_w) &= \sum_{i=1}^{k} w_i E(\bar{y}_i) \\ &= \sum_{i=1}^{k} w_i \bar{Y}_i \\ &= \sum_{i=1}^{k} \frac{N_i}{N} \bar{Y}_i \\ &= \frac{1}{N} \sum_{i=1}^{k} \sum_{j=1}^{N_i} Y_{ij} \\ &= \bar{Y} \end{aligned}$$

which completes the proof.

Next suppose we are interested in knowing the precision involved in using $\bar{y}_w$ then we require $\text{var}(\bar{y}_w)$

$$\begin{aligned} \text{var}(\bar{y}_w) &= \text{var}\left(\sum_{i=1}^{k} w_i \bar{y}_i\right) \\ &= \sum_{i=1}^{k} w_i^2 \, \text{var}(\bar{y}_i) \\ &= \sum_{i=1}^{k} w_i^2 \left(\frac{N_i - n_i}{N_i} \cdot \frac{S_i^2}{n_i}\right) \\ &= \frac{1}{N^2} \sum_{i=1}^{k} N_i^2 \left(\frac{N_i - n_i}{N_i}\right) \frac{S_i^2}{n_i} \end{aligned}$$

$$= \frac{1}{N^2} \sum_{i=1}^{k} N_i(N_i - n_i) \frac{S_i^2}{n_i}$$

and we estimate S_i^2 by s_i^2.

We note that $\text{Var}(\bar{y}_w)$ depends on S_i^2 which is the variance of the ith stratum the implication of this is that the more homogenous the strata within the greater the precision of the weighted mean.

3.3 Allocation of Sample Sizes in Stratified Sampling

It has been seen that the sample sizes contributes to the overall precision when estimating the population mean of stratified sampling because of this, procedures of allocating sample sizes must be statistically sound. We have established that the variance $\text{var}(\bar{y}_w) = \sum_{i=1}^{k} w_i^2 \left(\frac{N_i - n_i}{N_i} \right) \frac{S_i^2}{n_i}$ which depends on n_i, usually n_i can be fixed by the sampler at will, but there exist two main ways of fixing this sample size.

1. Proportional allocation
2. Optimal allocation

3.3.1 Proportional Allocation

Under this allocation scheme the idea is to allocate the sample in such a way that the sample fraction for each stratum is constant, i.e $f_i = \frac{n_i}{N_i}$ is constant where $i = 1, 2, \ldots, k$ this implies that

$$\frac{n_1}{N_1} = \frac{n_2}{N_2} = \cdots = \frac{n_k}{N_k} = \text{constant}$$
$$n_i = cN_i$$

and accordingly

$$\sum_{i=1}^{k} n_i = c \sum_{i=1}^{k} N_i$$
$$n = cN$$

thus $c = \frac{n}{N}$ this is usual sampling fraction. So it is clear that $n_i \propto N_i$ and hence the name proportional allocation. In other words each stratum is represented in the sample in proportion to its size.

3.3.2 Comparing SRS with Stratified Random Sampling under Proportional Allocation Scheme

We have that

$$\text{var}(\bar{y}_w)_{prop} = \left(\frac{1}{n} - \frac{1}{N} \right) \sum_{i=1}^{k} w_i S_i^2$$

and

$$\text{var}(\bar{y})_{SRSWOR} = \left(\frac{1}{n} - \frac{1}{N} \right) S^2$$

Now from the first principles, in a stratified scheme the overall variance should be computed as follows.

$$S^2 = \frac{1}{N-1} \sum_{i=1}^{N} \sum_{j=1}^{k} (Y_{ij} - \bar{Y})^2$$

$$= \frac{1}{N-1} \sum_{i=1}^{N} \sum_{j=1}^{k} (Y_{ij} - \bar{Y}_i + \bar{Y}_i - \bar{Y})^2$$

$$= \frac{1}{N-1} \sum_{i=1}^{N} \sum_{j=1}^{k} \left[(Y_{ij} - \bar{Y}_i)^2 + 2(Y_{ij} - \bar{Y}_i)(\bar{Y}_i - \bar{Y}) + (\bar{Y}_i - \bar{Y})^2 \right]$$

$$= \frac{1}{N-1} \left[\sum_{i=1}^{N} \sum_{j=1}^{k} (Y_{ij} - \bar{Y}_i)^2 + \sum_{i=1}^{N} N_i (\bar{Y}_i - \bar{Y})^2 \right]$$

The cross product term vanishes because $\sum_j (Y_{ij} - \bar{Y}_i) = 0$.
Thus:

$$S^2 = \frac{1}{N-1} \left[\sum_{i=1}^{k} (N_i - 1) S_i^2 + \sum_{i=1}^{k} N_i (\bar{Y}_i - \bar{Y})^2 \right]$$

If the strata means are all equal, then the second term vanishes and we have

$$S^2 \approx \sum_{i=1}^{k} \frac{N_i - 1}{N - 1} S_i^2$$

For large N, this is approximately $\sum w_i S_i^2$. In this case,

$$\text{var}(\bar{y}_w)_{prop} \approx \left(\frac{1}{n} - \frac{1}{N} \right) S^2 = \text{var}(\bar{y})_{SRSWOR}$$

If the strata means differ, the second term is positive and $S^2 > \sum w_i S_i^2$, so

$$\text{var}(\bar{y}_w)_{prop} < \text{var}(\bar{y})_{SRSWOR}$$

Thus stratified sampling with proportional allocation is never less precise than SRSWOR, and is strictly more precise unless the stratum means are all equal.

3.3.3 Neyman Allocation (Optimal Allocation for Fixed Total Sample Size)

We wish to minimize

$$\text{var}(\bar{y}_w) = \frac{1}{N^2} \sum_{i=1}^{k} N_i (N_i - n_i) \frac{S_i^2}{n_i}$$

subject to the condition that $\sum_{i=1}^{k} n_i = n$ (fixed total sample size)

We use Lagrange multiplier method. Define:

$$\phi = \text{var}(\bar{y}_w) + \lambda \left(\sum_{i=1}^{k} n_i - n \right)$$

where λ is the Lagrange multiplier.

For the minimum:

$$\frac{\partial \phi}{\partial n_i} = 0$$

$$\frac{\partial \phi}{\partial n_i} = \frac{\partial}{\partial n_i} \left\{ \frac{1}{N^2} \sum_{i=1}^{k} N_i (N_i - n_i) \frac{S_i^2}{n_i} + \lambda \left(\sum_{i=1}^{k} n_i - n \right) \right\}$$

$$= \frac{1}{N^2}\left(-\frac{N_i^2 S_i^2}{n_i^2}\right) + \lambda$$

$$\Rightarrow -\frac{w_i^2 S_i^2}{n_i^2} + \lambda = 0$$

$$\Rightarrow n_i^2 = \frac{w_i^2 S_i^2}{\lambda}$$

$$\Rightarrow n_i = \frac{w_i S_i}{\sqrt{\lambda}}$$

We observe that $\frac{\partial^2 \phi}{\partial n_i^2} = \frac{2w_i^2 S_i^2}{n_i^3} > 0$, which means this minimizes $\text{var}(\bar{y}_w)$.

To find λ, sum both sides:

$$\sum_{i=1}^{k} n_i = \sum_{i=1}^{k} \frac{w_i S_i}{\sqrt{\lambda}}$$

$$n = \frac{1}{\sqrt{\lambda}} \sum_{i=1}^{k} w_i S_i$$

$$\sqrt{\lambda} = \frac{\sum_{i=1}^{k} w_i S_i}{n}$$

Substituting back:

$$n_i = \frac{w_i S_i}{\sqrt{\lambda}} = \frac{w_i S_i}{\frac{\sum_{i=1}^{k} w_i S_i}{n}} = \frac{n(w_i S_i)}{\sum_{i=1}^{k} w_i S_i}$$

Thus the optimal allocation (Neyman allocation) is:

$$n_i \propto N_i S_i$$

Then we can conclude that Neyman allocation is more efficient strategy than proportional allocation.

3.3.4 Example

A sample of 30 students is to be drawn from a population of 300 students belonging to two colleges A and B. The means and standard deviations of their marks are given below:

	Total number of students	$\bar{y}_i$	S_i
College A	200	30	10
College B	100	60	40

Use the information to confirm that Neyman's allocation scheme is a more efficient scheme when compared to proportional allocation.

Solution:

For proportional allocation:

$$n_A = \frac{30}{300} \times 200 = 20$$

$$n_B = \frac{30}{300} \times 100 = 10$$

$$\text{var}(\bar{y}_w)_{prop} = \sum_{i=1}^{2} w_i^2 \left(\frac{N_i - n_i}{N_i}\right) \frac{S_i^2}{n_i}$$

$$w_A = \frac{200}{300} = \frac{2}{3}, w_B = \frac{100}{300} = \frac{1}{3}$$

$$\begin{aligned}
\text{var}(\bar{y}_w)_{prop} &= \left(\frac{2}{3}\right)^2 \left(\frac{200-20}{200}\right) \frac{10^2}{20} + \left(\frac{1}{3}\right)^2 \left(\frac{100-10}{100}\right) \frac{40^2}{10} \\
&= \frac{4}{9} \times \frac{180}{200} \times \frac{100}{20} + \frac{1}{9} \times \frac{90}{100} \times \frac{1600}{10} \\
&= \frac{4}{9} \times 0.9 \times 5 + \frac{1}{9} \times 0.9 \times 160 \\
&= \frac{4}{9} \times 4.5 + \frac{1}{9} \times 144 \\
&= \frac{18}{9} + \frac{144}{9} = \frac{162}{9} = 18
\end{aligned}$$

For Neyman allocation:

$$n_A = \frac{n(N_A S_A)}{N_A S_A + N_B S_B} = \frac{30 \times (200 \times 10)}{200 \times 10 + 100 \times 40} = \frac{30 \times 2000}{2000 + 4000} = \frac{60000}{6000} = 10$$

$$n_B = 30 - 10 = 20$$

$$\begin{aligned}
\text{var}(\bar{y}_w)_{Neyman} &= \left(\frac{2}{3}\right)^2 \left(\frac{200-10}{200}\right) \frac{10^2}{10} + \left(\frac{1}{3}\right)^2 \left(\frac{100-20}{100}\right) \frac{40^2}{20} \\
&= \frac{4}{9} \times \frac{190}{200} \times \frac{100}{10} + \frac{1}{9} \times \frac{80}{100} \times \frac{1600}{20} \\
&= \frac{4}{9} \times 0.95 \times 10 + \frac{1}{9} \times 0.8 \times 80 \\
&= \frac{4}{9} \times 9.5 + \frac{1}{9} \times 64 \\
&= \frac{38}{9} + \frac{64}{9} = \frac{102}{9} = 11.33
\end{aligned}$$

Since $11.33 < 18$, Neyman allocation is more efficient than proportional allocation.

3.3.5 Exercise

Investigate between $\text{var}(\bar{y}_w)_{Neyman}$ and $\text{var}(\bar{y}_w)_{SRSWOR}$

3.3.6 Minimizing Variance for a Fixed Cost

Here we consider the fixed cost to be $C = a + \sum_{i=1}^{k} c_i n_i$ where a is overhead cost and c_i is the cost per unit in stratum i. We require the minimum variance such that $C = a + \sum_{i=1}^{k} c_i n_i$ is fixed. Convert this conditional minimization case to an unconditional case so that we write:

$$\phi = \text{var}(\bar{y}_w) + \mu \left(a + \sum_{i=1}^{k} c_i n_i - C \right)$$

where μ is a Lagrange multiplier. So we minimize unconditionally:

$$\phi = \sum_{i=1}^{k} \left(\frac{N_i}{N}\right)^2 \frac{N_i - n_i}{N_i} \frac{S_i^2}{n_i} + \mu \left(a + \sum_{i=1}^{k} c_i n_i - C\right)$$

$$= \sum_{i=1}^{k} \frac{w_i^2 S_i^2}{n_i} - \sum_{i=1}^{k} \frac{w_i^2 S_i^2}{N_i} + \mu a + \mu \sum_{i=1}^{k} c_i n_i - \mu C$$

Differentiating with respect to n_i:

$$\frac{\partial \phi}{\partial n_i} = -\frac{w_i^2 S_i^2}{n_i^2} + \mu c_i = 0$$

$$\Rightarrow \frac{w_i^2 S_i^2}{n_i^2} = \mu c_i$$

$$\Rightarrow n_i^2 = \frac{w_i^2 S_i^2}{\mu c_i}$$

$$\Rightarrow n_i = \frac{w_i S_i}{\sqrt{\mu c_i}}$$

Now sum over all strata:

$$\sum_{i=1}^{k} n_i = \sum_{i=1}^{k} \frac{w_i S_i}{\sqrt{\mu c_i}}$$

$$n = \frac{1}{\sqrt{\mu}} \sum_{i=1}^{k} \frac{w_i S_i}{\sqrt{c_i}}$$

$$\sqrt{\mu} = \frac{1}{n} \sum_{i=1}^{k} \frac{w_i S_i}{\sqrt{c_i}}$$

Substituting back:

$$n_i = \frac{w_i S_i}{\sqrt{c_i}} \cdot \frac{n}{\sum_{i=1}^{k} \frac{w_i S_i}{\sqrt{c_i}}}$$

$$= \frac{n \cdot \frac{N_i S_i}{\sqrt{c_i}}}{\sum_{i=1}^{k} \frac{N_i S_i}{\sqrt{c_i}}}$$

meaning that $n_i \propto \frac{N_i S_i}{\sqrt{c_i}}$

- The larger the size of the stratum the larger should be the size of the sample selected from it.
- The larger the variability within the stratum the larger should be the size of the sample selected from that stratum.
- The cheaper the cost of sampling within the ith stratum the larger should be the size of the sample from that stratum.

3.3.7 Example

A stratified population has 5 strata. The stratum sizes N_i and means $\bar{Y}_i$ and S_i^2 of some variable Y are as follows;

Stratum	N_i	$\bar{Y}_i$	S_i^2
1	117	7.3	1.31
2	98	6.9	2.03
3	74	11.2	1.13
4	41	9.1	1.96
5	45	9.6	1.74

Calculate the overall population mean and variance. For a stratified simple random sample of size 80, determine the appropriate stratum sample sizes under Proportional allocation and Neyman allocation.

Solution:

$$N = \sum_{h=1}^{5} N_h = 117 + 98 + 74 + 41 + 45 = 375$$

$$w_h = \frac{N_h}{N}$$

$$w_1 = \frac{117}{375}, w_2 = \frac{98}{375}, w_3 = \frac{74}{375}, w_4 = \frac{41}{375}, w_5 = \frac{45}{375}$$

Overall population mean:

$$\bar{Y} = \sum_{h=1}^{5} w_h \bar{Y}_h$$

$$= \frac{117}{375} \times 7.3 + \frac{98}{375} \times 6.9 + \frac{74}{375} \times 11.2 + \frac{41}{375} \times 9.1 + \frac{45}{375} \times 9.6$$

$$= \frac{854.1 + 676.2 + 828.8 + 373.1 + 432}{375}$$

$$= \frac{3164.2}{375} = 8.438$$

For a sample of size 80 under proportional allocation:

$$n_h = \frac{80}{375} N_h$$

$$n_1 = \frac{80}{375} \times 117 = 24.96 \approx 25$$

$$n_2 = \frac{80}{375} \times 98 = 20.9 \approx 21$$

$$n_3 = \frac{80}{375} \times 74 = 15.78 \approx 16$$

$$n_4 = \frac{80}{375} \times 41 = 8.74 \approx 9$$

$$n_5 = \frac{80}{375} \times 45 = 9.6 \approx 10$$

Check: $25 + 21 + 16 + 9 + 10 = 81$ (slight adjustment needed)

Under Neyman allocation:

$$n_h = \frac{nN_hS_h}{\sum_{h=1}^{5} N_hS_h}$$

First compute N_hS_h:

$$S_h = \sqrt{S_h^2}$$

$$S_1 = \sqrt{1.31} = 1.145, \quad N_1S_1 = 117 \times 1.145 = 133.97$$

$$S_2 = \sqrt{2.03} = 1.425, \quad N_2S_2 = 98 \times 1.425 = 139.65$$

$$S_3 = \sqrt{1.13} = 1.063, \quad N_3S_3 = 74 \times 1.063 = 78.66$$

$$S_4 = \sqrt{1.96} = 1.4, \quad N_4S_4 = 41 \times 1.4 = 57.4$$

$$S_5 = \sqrt{1.74} = 1.319, \quad N_5S_5 = 45 \times 1.319 = 59.36$$

$$\sum N_hS_h = 133.97 + 139.65 + 78.66 + 57.4 + 59.36 = 469.04$$

Then:

$$n_1 = \frac{80 \times 133.97}{469.04} = 22.85 \approx 23$$

$$n_2 = \frac{80 \times 139.65}{469.04} = 23.82 \approx 24$$

$$n_3 = \frac{80 \times 78.66}{469.04} = 13.42 \approx 13$$

$$n_4 = \frac{80 \times 57.4}{469.04} = 9.79 \approx 10$$

$$n_5 = \frac{80 \times 59.36}{469.04} = 10.12 \approx 10$$

Check: $23 + 24 + 13 + 10 + 10 = 80$

3.3.8 Exercise

All the farms in a country are stratified by farm size and mean number of hectares of wheat per farm in each stratum, with the following results.

Farm size (hectares)	No. of farms	Mean wheat (hectares)	standard deviation
0-20	368	2.7	2.1
21-40	425	8.1	3.6
41-60	389	12.1	3.9
61-80	316	16.9	5.1
81-100	174	20.8	6.1
101-120	98	25.2	6.5
121+	138	31.8	9.1

For a sample of 100 farms, compute the sizes in each stratum under stratified simple random sampling with:

1. Proportional allocation
2. Neyman allocation

Solution : Proportional Allocation

$$n = 100, \quad n_h = \left(\frac{n}{N}\right) N_h$$

$$N = \sum_{h=1}^{7} N_h = 368 + 425 + 389 + 316 + 174 + 98 + 138 = 1908$$

$$n_1 = \left(\frac{100}{1908}\right) 368 = 19.287 \approx 19.3$$

$$n_2 = \left(\frac{100}{1908}\right) 425 = 22.275 \approx 22.3$$

$$n_3 = \left(\frac{100}{1908}\right) 389 = 20.388 \approx 20.4$$

$$n_4 = \left(\frac{100}{1908}\right) 316 = 16.562 \approx 16.6$$

$$n_5 = \left(\frac{100}{1908}\right) 174 = 9.119 \approx 9.1$$

$$n_6 = \left(\frac{100}{1908}\right) 98 = 5.136 \approx 5.1$$

$$n_7 = \left(\frac{100}{1908}\right) 138 = 7.233 \approx 7.2$$

Check sum: $19.3 + 22.3 + 20.4 + 16.6 + 9.1 + 5.1 + 7.2 = 100$

Solution : Neyman Allocation

$$n_h = \frac{nN_hS_h}{\sum_{h=1}^{7} N_hS_h}$$

First compute N_hS_h:

$$\begin{aligned}
N_1S_1 &= 368 \times 2.1 = 772.8 \\
N_2S_2 &= 425 \times 3.6 = 1530.0 \\
N_3S_3 &= 389 \times 3.9 = 1517.1 \\
N_4S_4 &= 316 \times 5.1 = 1611.6 \\
N_5S_5 &= 174 \times 6.1 = 1061.4 \\
N_6S_6 &= 98 \times 6.5 = 637.0 \\
N_7S_7 &= 138 \times 9.1 = 1255.8
\end{aligned}$$

$$\sum N_hS_h = 772.8 + 1530.0 + 1517.1 + 1611.6 + 1061.4 + 637.0 + 1255.8 = 8385.7$$

Then:

$$n_1 = \frac{100 \times 772.8}{8385.7} = \frac{77280}{8385.7} = 9.215 \approx 9.2$$
$$n_2 = \frac{100 \times 1530.0}{8385.7} = \frac{153000}{8385.7} = 18.245 \approx 18.2$$
$$n_3 = \frac{100 \times 1517.1}{8385.7} = \frac{151710}{8385.7} = 18.091 \approx 18.1$$
$$n_4 = \frac{100 \times 1611.6}{8385.7} = \frac{161160}{8385.7} = 19.218 \approx 19.2$$
$$n_5 = \frac{100 \times 1061.4}{8385.7} = \frac{106140}{8385.7} = 12.657 \approx 12.7$$
$$n_6 = \frac{100 \times 637.0}{8385.7} = \frac{63700}{8385.7} = 7.596 \approx 7.6$$
$$n_7 = \frac{100 \times 1255.8}{8385.7} = \frac{125580}{8385.7} = 14.975 \approx 15.0$$

Check sum: $9.2 + 18.2 + 18.1 + 19.2 + 12.7 + 7.6 + 15.0 = 100$

4 Ratio Method of Estimation

In developing the theory of sample surveys most cases have considered only estimates based on simple averages of sample values. There are other methods however which make use of auxiliary information and which under certain situations give more reliable estimates of the population parameters.

One of such methods is the ratio method of estimation which forms a basis for all other methods that use auxiliary information.

Y_i is the survey measurement for the ith unit of the population. X_i is the value of the auxiliary information or measurement for the ith unit $(i = 1, 2, \ldots, N)$.

Note X_i are assumed known for all the units in the population. y_i is the measurement for the ith unit in the sample. x_i is the value of the auxiliary information for the ith unit in the sample.

$$r_i = \frac{y_i}{x_i} \Rightarrow \bar{r} = \frac{1}{n}\sum_{i=1}^{n}\frac{y_i}{x_i} = \frac{1}{n}\sum_{i=1}^{n} r_i$$
$$R_i = \frac{Y_i}{X_i}$$

The population ratio is defined as:

$$R = \frac{\bar{Y}}{\bar{X}} = \frac{\sum_{i=1}^{N} Y_i}{\sum_{i=1}^{N} X_i}$$

The sample ratio is:

$$\hat{R} = \frac{\bar{y}}{\bar{x}} = \frac{\sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} x_i}$$

The ratio estimator of the population mean $\bar{Y}$ is:

$$\bar{y}_R = \hat{R}\bar{X} = \frac{\bar{y}}{\bar{x}}\bar{X}$$

And the ratio estimator of the population total Y is:

$$\hat{Y}_R = \hat{R}X = \frac{\bar{y}}{\bar{x}}X$$

4.1 Bias of the Ratio Estimator

The ratio estimator is biased in general. We can approximate the bias.

Let

$$\delta_y = \frac{\bar{y} - \bar{Y}}{\bar{Y}}, \quad \delta_x = \frac{\bar{x} - \bar{X}}{\bar{X}}$$

Then

$$\hat{R} = \frac{\bar{y}}{\bar{x}} = \frac{\bar{Y}(1+\delta_y)}{\bar{X}(1+\delta_x)} = R(1+\delta_y)(1+\delta_x)^{-1}$$

Assuming $|\delta_x| < 1$, we can expand:

$$(1+\delta_x)^{-1} = 1 - \delta_x + \delta_x^2 - \delta_x^3 + \cdots$$

Thus:

$$\begin{aligned}\hat{R} &= R(1+\delta_y)(1-\delta_x+\delta_x^2-\cdots) \\ &= R(1+\delta_y-\delta_x-\delta_y\delta_x+\delta_x^2+\cdots)\end{aligned}$$

Taking expectation and noting that $E(\delta_y) = E(\delta_x) = 0$ in SRS:

$$E(\hat{R}) \approx R\left[1 - \frac{\mathrm{Cov}(\bar{x},\bar{y})}{\bar{X}\bar{Y}} + \frac{\mathrm{Var}(\bar{x})}{\bar{X}^2}\right]$$

More precisely, to first order of approximation:

$$E(\hat{R}) \approx R\left(1 - \frac{1-f}{n}\frac{S_{xy}}{\bar{X}\bar{Y}} + \frac{1-f}{n}\frac{S_x^2}{\bar{X}^2}\right)$$

The bias of $\hat{R}$ as an estimator of R is approximately:

$$\mathrm{Bias}(\hat{R}) \approx -\frac{1-f}{n\bar{X}^2}(RS_x^2 - S_{xy})$$

For the ratio estimator of the mean $\bar{y}_R = \hat{R}\bar{X}$:

$$E(\bar{y}_R) \approx \bar{Y} - \frac{1-f}{n\bar{X}}(RS_x^2 - S_{xy})$$

The bias is small if the sample size is large or if $R \approx \frac{S_{xy}}{S_x^2}$, i.e., if the regression of Y on X passes through the origin.

4.2 Variance of the Ratio Estimator

To the first order of approximation, the variance of the ratio estimator is:

$$\begin{aligned}\mathrm{Var}(\bar{y}_R) &\approx \frac{1-f}{n}(S_y^2 - 2RS_{xy} + R^2S_x^2) \\ &= \frac{1-f}{n}\cdot\frac{1}{N-1}\sum_{i=1}^{N}(Y_i - RX_i)^2\end{aligned}$$

Or in terms of coefficient of variation:

$$\mathrm{Var}(\bar{y}_R) \approx \frac{1-f}{n}\bar{Y}^2(C_y^2 + C_x^2 - 2\rho C_xC_y)$$

where $C_y = \frac{S_y}{\bar{Y}}$, $C_x = \frac{S_x}{\bar{X}}$, and ρ is the correlation coefficient between Y and X.

The estimated variance (from sample) is:

$$\begin{aligned}\widehat{\mathrm{Var}}(\bar{y}_R) &= \frac{1-f}{n}\cdot\frac{1}{n-1}\sum_{i=1}^{n}(y_i - \hat{R}x_i)^2 \\ &= \frac{1-f}{n(n-1)}\left(\sum y_i^2 - 2\hat{R}\sum x_iy_i + \hat{R}^2\sum x_i^2\right)\end{aligned}$$

4.3 Example

Suppose that the following summarized information is made available. $n = 25$, $N = 275$, $\bar{x} = 9.2$, $\bar{y} = 2.6$, $\sum_{i=1}^{25} x_i^2 = 2200$, $\sum_{i=1}^{25} x_i y_i = 500$, $\sum_{i=1}^{25} y_i^2 = 170$. Estimate R and $\text{var}(\hat{R})$.

Solution:

$$\hat{R} = \frac{\bar{y}}{\bar{x}} = \frac{2.6}{9.2} = 0.2826$$

$$\widehat{\text{Var}}(\hat{R}) = \frac{1-f}{n\bar{x}^2} \cdot \frac{1}{n-1} \sum (y_i - \hat{R}x_i)^2$$

First compute $\sum (y_i - \hat{R}x_i)^2$:

$$\sum y_i^2 - 2\hat{R} \sum x_i y_i + \hat{R}^2 \sum x_i^2$$

$$= 170 - 2(0.2826)(500) + (0.2826)^2(2200)$$

$$= 170 - 282.6 + (0.07986)(2200)$$

$$= 170 - 282.6 + 175.69 = 63.09$$

$$\frac{1}{n-1} \sum (y_i - \hat{R}x_i)^2 = \frac{63.09}{24} = 2.62875$$

$$1 - f = 1 - \frac{25}{275} = 1 - 0.0909 = 0.9091$$

$$\widehat{\text{Var}}(\hat{R}) = \frac{0.9091}{25 \times (9.2)^2} \times 2.62875$$

$$= \frac{0.9091}{25 \times 84.64} \times 2.62875$$

$$= \frac{0.9091}{2116} \times 2.62875$$

$$= 0.0004296 \times 2.62875 = 0.001129$$

The standard error of $\hat{R}$ is $\sqrt{0.001129} = 0.0336$

4.3.1 Exercise

The following results were obtained from a sample of 48 stores:

$$\bar{x} = 11.41, \bar{y} = 12.07, \sum x_i^2 = 8431.7, \sum x_i y_i = 8564.1, \sum y_i^2 = 9270.6$$

Estimate the sampling variance of the ratio estimator and the 95% confidence interval for R.

Solution

Given: $\bar{x} = 11.41, \bar{y} = 12.07, \sum x_i^2 = 8431.7, \sum x_i y_i = 8564.1, \sum y_i^2 = 9270.6, n = 48$

$$\hat{R} = \frac{\bar{y}}{\bar{x}} = \frac{12.07}{11.41} = 1.0578 \approx 1.06$$

The approximate sampling variance of the estimator is:

$$\widehat{\text{var}}(\hat{R}) = \frac{1-f}{n\bar{x}^2} \cdot \frac{1}{n-1} \sum (y_i - \hat{R}x_i)^2$$

First compute $\sum (y_i - \hat{R}x_i)^2$:

$$\sum y_i^2 - 2\hat{R} \sum x_i y_i + \hat{R}^2 \sum x_i^2$$

$$= 9270.6 - 2(1.0578)(8564.1) + (1.0578)^2(8431.7)$$

$$= 9270.6 - 18118.5 + 9430.3 = 582.4$$

$$\frac{1}{n-1} \sum (y_i - \hat{R}x_i)^2 = \frac{582.4}{47} = 12.3915$$

Assuming the population is large so that $1 - f \approx 1$:

$$\widehat{\text{var}}(\hat{R}) = \frac{12.3915}{48 \times (11.41)^2} = \frac{12.3915}{48 \times 130.1881} = \frac{12.3915}{6249.03} = 0.001983$$

$$s.e.(\hat{R}) = \sqrt{0.001983} = 0.0445$$

The 95% confidence interval for R is:

$$\hat{R} \pm 1.96 \times 0.0445 = 1.0578 \pm 0.0872$$

$$0.9706 < R < 1.1450$$

Average food price increase would be unwise in light of this approximate interval (the wide range reflects small sample size in survey samples).

4.3.2 Exercise

Under what circumstances does the ratio estimation procedure fail? In the event that some auxiliary information is missing how would one handle ratio estimation? Suppose that in the survey characteristics there are some missing values, how would you handle the situation?

4.3.3 Exercise

Obtain the approximate variance of ratio estimation of the population mean $\bar{Y}$ of a characteristic Y under study with auxiliary variable X.

4.4 The Optimality of Ratio Estimator under Superpopulation Model

Assume that $y_i's(i = 1, 2, \ldots, N)$ are realised values of a sample drawn from a superpopulation model of N independent random variables $Y_i's(i = 1, 2, \ldots, N)$ having a joint probability distribution ξ such that $E_\xi(Y_i) = \beta X_i$, $\text{var}_\xi(Y_i) = \sigma^2 \text{var}(X_i)$ and $\text{cov}(Y_i, Y_j) = 0$ where $i \neq j$, $\text{var}(X_i) > 0$, $X_i > 0$, β and σ^2 are known constants.

So far we have assumed that the sample is a random variable but the survey measurements $y_i's$ are fixed constants. We now take the opposite view i.e the sample is fixed, while the $y_i's, (i = 1, 2, \ldots, N)$ are realised values of the random variable Y_i.

We note that $T = \sum_{i=1}^{N} y_i = \sum_{i \in s} y_i + \sum_{i \in r} y_i$ where s is the sample, r is non sample, $\sum_s$ is summation over the sample units and $\sum_r$ is the summation over the non sample units. Hence the problem of estimating T is essentially a problem of predicting the sum of unobserved random variables i.e $\sum_r Y_i$. From the sample we infer about ξ which is used to predict $\sum_r Y_i$. The best predictor of $\sum_r y_i$ is;

$$\hat{E}\left(\sum_r Y_i\right) = \sum_r \beta X_i = \hat{\beta} \sum_r X_i$$

where $\hat{\beta}$ is the best estimator of β.

For a given sample, an estimator $\hat{T}$ is said to be unbiased of T under a superpopulation model ξ if $E_\xi\left(\hat{T} - T\right) = 0$.

Note: $E(y_i) = \beta x_i \Rightarrow y_i = \beta x_i + e_i$. Where e_i is such that $\text{var}(e_i) = \sigma^2 \text{var}(x_i)$ and $\text{cov}(e_i e_j) = 0$, $i \neq j$ clearly:

$$s = \sum_s \frac{e_i^2}{\text{var}(x_i)} = \sum_s \frac{1}{\text{var}(x_i)} (y_i - \beta x_i)^2$$

Now find β that minimises s i.e

$$\frac{\partial s}{\partial \beta} = -\sum_s \frac{x_i}{\text{var}(x_i)} (y_i - \beta x_i) = 0$$

$$\Rightarrow \beta \sum_s \frac{x_i^2}{\text{var}(x_i)} = \sum_s \frac{x_i y_i}{\text{var}(x_i)}$$

$$\hat{\beta} = \frac{\sum_s \frac{x_i y_i}{\text{var}(x_i)}}{\sum_s \frac{x_i^2}{\text{var}(x_i)}}$$

Hence the BLUE of T is

$$\hat{T}_{BLUE} = \sum_s y_i + \hat{\beta} \sum_r x_i$$

Which was obtained by Royall 1970 - Biometrica. We present this result as a theorem: under the superpopulation probability model ξ and for any sampling design P_s , if $\hat{T}$ is a linear estimator of $T = \sum_{i=1}^{N} y_i$ such that $E_\xi\left(\hat{T} - T\right) = 0$ then for every s for which $P_s > 0$ the BLUE of T is:

$$\hat{T} = \sum_{i \in s} y_i + \hat{\beta} \sum_{i \in r} x_i$$

Where

$$\hat{\beta} = \frac{\sum_{i \in s} \frac{x_i y_i}{\text{var}(x_i)}}{\sum_{i \in s} \frac{x_i^2}{\text{var}(x_i)}}$$

and the expected mean square error is:

$$E_\xi\left[\text{MSE}\left(\hat{T}\right)\right] = \sigma^2\left\{\sum_{s\in S} P_s \frac{\left(\sum_{i\in r} x_i\right)^2}{\sum_{i\in s}\frac{x_i^2}{\text{var}(x_i)}} + \sum_{i\in r}\text{var}(x_i)\right\}$$

4.4.1 SPECIAL CASE: $\text{var}(x_i) = x_i$

Then

$$\hat{\beta} = \frac{\sum_{i\in s}\frac{x_i y_i}{\text{var}(x_i)}}{\sum_{i\in s}\frac{x_i^2}{\text{var}(x_i)}} = \frac{\sum_{i\in s} y_i}{\sum_{i\in s} x_i}$$

Therefore

$$\begin{aligned}\hat{T}_{BLUE} &= \sum_s y_i + \left(\frac{\sum_{i\in s} y_i}{\sum_{i\in s} x_i}\right)\sum_{i\in r} x_i \\ &= \frac{\sum_{i\in s} y_i}{\sum_{i\in s} x_i}\left(\sum_{i\in s} x_i + \sum_{i\in r} x_i\right) \\ &= \frac{\sum_{i\in s} y_i}{\sum_{i\in s} x_i}\sum_{i=1}^{N} x_i\end{aligned}$$

Which is the ratio estimator of the population total T. Hence the ratio estimator is the BLUE under the superpopulation model ξ when $\text{var}_\xi(y_i) = \sigma^2 x_i$.

In this special case;

$$\begin{aligned}E_\xi\left[\text{MSE}\left(\hat{T}\right)\right] &= \sigma^2\left\{\sum_{s\in S} P_s\left[\frac{\left(\sum_{i\in r} x_i\right)^2}{\sum_{i\in s}\left(\frac{x_i^2}{x_i}\right)} + \sum_{i\in r} x_i\right]\right\} \\ &= \sigma^2\left\{\sum_{s\in S} P_s\left[\frac{\left(\sum_{i\in r} x_i\right)^2}{\sum_{i\in s} x_i} + \sum_{i\in r} x_i\right]\right\}\end{aligned}$$

From this expression, it is clear that an optimal sample is one for which $\sum_{i\in s} x_i$ attains the maximum value. This is simply the sample composed of the n units whose x values are largest.

Under SRSWOR $P_s = \frac{1}{{}^N C_n}$, but more conveniently:

$$E_\xi\left[\text{MSE}\left(\hat{T}\right)\right] = \sigma^2\left\{\frac{1}{{}^N C_n}\sum_{s\in S}\left[\frac{\left(\sum_{i\in r} x_i\right)^2}{\sum_{i\in s} x_i} + \sum_{i\in r} x_i\right]\right\}$$

So that

$$\begin{aligned}E_\xi\left[\text{MSE}\left(\bar{y}_R\right)\right] &= \frac{(N-n)\bar{x}\bar{x}_r}{N\bar{x}_s}\sigma^2 \\ &= (1-f)\frac{\bar{x}\bar{x}_r}{\bar{x}_s}\sigma^2\end{aligned}$$

where $\bar{x}_r = \frac{1}{N-n}\sum_{i\in r} x_i$

Several questions concerning the robustness of the ratio estimator have been investigated by Royall & Herson (1973 a,b) in J.A.S.A. Clearly, the ratio estimator is unbiased under the superpopulation model ξ, whatever the variance functions $\text{var}(x_i)$.

Note: Robustness means that the parameter in question remains insensitive to all changes or misspecifications in the model.

Suppose that the regression line of Y on X does not pass through the origin. Is the estimator still unbiased? i.e $E(y_i) = \beta x_i + \alpha$.

Now

$$\begin{aligned} E_\xi(\bar{y}_R) &= \frac{\bar{X}}{\bar{x}} E_\xi(\bar{y}) \\ &= \frac{\bar{X}}{\bar{x}} \cdot \frac{1}{n} \sum_{i \in s} E_\xi(y_i) \\ &= \frac{\bar{X}}{n\bar{x}} \sum_{i \in s} (\alpha + \beta x_i) \\ &= \frac{\bar{X}}{n\bar{x}} (n\alpha + \beta \sum_{i \in s} x_i) \\ &= \frac{\alpha \bar{X}}{\bar{x}} + \beta \bar{X} \end{aligned}$$

Next

$$\begin{aligned} E_\xi(\bar{Y}) &= E_\xi \left(\frac{1}{N} \sum_{i=1}^{N} Y_i \right) \\ &= \frac{1}{N} \sum_{i=1}^{N} E_\xi(Y_i) \\ &= \frac{1}{N} \sum_{i=1}^{N} (\alpha + \beta x_i) \\ &= \alpha + \beta \bar{X} \end{aligned}$$

Hence $E(\bar{y}_R) \neq E(\bar{Y})$

Therefore, it is that now the ratio estimator is biased with

$$\text{Bias} = \frac{\alpha \bar{X}}{\bar{x}} - \alpha = \alpha \left(\frac{\bar{X} - \bar{x}}{\bar{x}} \right)$$

This bias disappears if $\bar{X} = \bar{x}$. A sample in which this holds is normally called a balanced sample. i.e (sample mean = population mean).

4.5 Estimation of the MSE of the Ratio Estimator

$$\begin{aligned} \text{MSE}(\bar{y}_R) &\approx \frac{N-n}{Nn} \left[S_y^2 - 2RS_{xy} + R^2 S_x^2 \right] \\ &\approx \frac{N-n}{Nn} \left[\frac{\sum_{i=1}^{N} (y_i - Rx_i)^2}{N-1} \right] \end{aligned}$$

An obvious estimator of MSE($\bar{y}_R$) is found by using the sample analogues of the above. i.e

$$\hat{V}_0 = \frac{N-n}{Nn} \left[\frac{\sum_{i=1}^{n} \left(y_i - \hat{R} x_i \right)^2}{n-1} \right]$$

$$= \frac{N-n}{Nn} \left[\frac{\left(\sum_{i=1}^{n} y_i^2 - 2\hat{R} \sum_{i=1}^{n} x_i y_i + \hat{R}^2 \sum_{i=1}^{n} x_i^2 \right)}{n-1} \right]$$

The other commonly used estimators of MSE($\bar{y}_R$) are:

4.5.1 Exercise

Part of coniferous forests 280 trees of the same species and of similar ages. A preliminary estimate is required of the total weight of timber that trees will yield. A forestry expert claims to be able to make fairly accurate assessments of the yield from any tree merely by visual inspection, and makes such assessments for all 280 trees. He assesses the total yield as 439.5 tonnes. Subsequently, 25 trees picked at random are felled and their timber yields y_i and their corresponding assessed yields, x_i , provide the following summary results.

$$\sum_{i=1}^{25} x_i^2 = 73.47, \quad \sum_{i=1}^{25} x_i y_i = 70.64, \quad \sum_{i=1}^{25} x_i = 41.4, \quad \sum_{i=1}^{25} y_i = 39.8$$

Estimate the total yield using either the ratio or regression estimator, whichever seems most appropriate. Compare the efficiencies of ratio estimator, the regression estimator, and the estimator based on the sample of y_i values alone.

5 Sampling with Varying Probabilities

So far we have assumed that samples are selected by SRSWOR or SRSWR in which case each unit in the population has an equal probability of being selected. If the units of the population vary considerably in size, SRS may not be appropriate, since it does not take into consideration the possible importance of larger units in the populations.

There are various ways of solving this problem, one is to assign unequal probability of selection to different units in the population i.e if the units vary in size and the variable under study is correlated with size, probabilities of selection may be assigned in proportion to the size of the unit. A selection procedure in which the units are selected with probabilities proportional to the measure of the size is known as sampling with PPS (Probability Proportional to Size).

5.1 Procedure of Selecting a Sample with Probability Proportional to Size (PPS)

To draw a sample of size n from a population of size N with PPS, we proceed as follows: Let x_i be an integer proportional to size of the ith unit $(i = 1, 2 \ldots, N)$. We form successive cumulative totals: $Y_1 = x_1$, $Y_2 = x_1 + x_2$, $Y_3 = x_1 + x_2 + x_3$, ... , $Y_N = x_1 + x_2 + x_3 + \cdots + x_N$. We then draw a random number R not exceeding $Y_N = \sum_{j=1}^{N} x_j$ from table of random numbers. If $Y_{i-1} < R \leq Y_i$, the ith unit is selected. The procedure is repeated until a sample size n is obtained (with replacement).

5.2 Example

Farms having: 10, 20, 30, 15, 25, 35, 65, 55, 50, 5 acres of land under maize. It is desired to select a sample of size 4 with replacement and with probability proportional to the area under maize. The total area under maize is 310. The first step is to form cumulative totals (C.T) and ranges as follows.

Farm	Size	Cumulative Total	Range
1	10	10	1-10
2	20	30	11-30
3	30	60	31-60
4	15	75	61-75
5	25	100	76-100
6	35	135	101-135
7	65	200	136-200
8	55	255	201-255
9	50	305	256-305
10	5	310	306-310

Solution

To select a farm, we choose a random number between 1 and 310 from random number tables. Suppose the chosen number is 66, we know that it falls in the range 61-75 and hence the 4th farm is selected. We draw 3 more random numbers and proceed in a similar way. Suppose the selected numbers are 102, 301 and 74 then the farms selected will be 6th, 9th and 4th. Hence the fourth (4th) farm is selected twice (since sampling is with replacement).

5.3 Estimation of the Population Mean ($\bar{Y}$)

Let Y_i be the value of Y for the ith unit in the population ($i = 1, 2, \ldots, N$) Let y_i be the value of y for the ith unit in the sample ($i = 1, 2, \ldots, n$) Let P_i be the probability of selecting the ith unit in the population (with replacement).

We now consider the problem of estimating the population mean $\bar{Y}$ based on a sample of n units, selected with probabilities P_i with replacement.

Define:

$$Z_i = \frac{Y_i}{NP_i}, \quad i = 1, \ldots, N$$

$$z_i = \frac{y_i}{NP_i}, \quad i = 1, \ldots, n$$

Consider $\bar{z} = \frac{1}{n}\sum_{i=1}^{n} z_i$ as an estimator of $\bar{Y}$.

Then

$$\begin{aligned} E(\bar{z}) &= \frac{1}{n}\sum_{i=1}^{n} E(z_i) \\ &= \frac{1}{n}\sum_{i=1}^{n} E\left(\frac{y_i}{NP_i}\right) \\ &= \frac{1}{n}\sum_{i=1}^{n}\left[\sum_{i=1}^{N}\left(\frac{Y_iP_i}{NP_i}\right)\right] \\ &= \frac{1}{n}\sum_{i=1}^{n}\frac{1}{N}\sum_{i=1}^{N} Y_i \\ &= \bar{Y} \end{aligned}$$

Thus $\bar{z}$ is an unbiased estimator of $\bar{Y}$.

The variance of z_i:

$$\text{var}(z_i) = E(z_i - E(z_i))^2$$

$$= \sum_{i=1}^{N} (z_i - \bar{Y})^2 P_i$$

$$= \sum_{i=1}^{N} \left\{ \left(\frac{Y_i}{NP_i} - \bar{Y} \right)^2 P_i \right\}$$

$$= \sigma_z^2$$

$$\begin{aligned} \text{var}(\bar{z}) &= E(\bar{z} - \bar{Y})^2 \\ &= \text{var}\left(\frac{1}{n} \sum_{i=1}^{n} z_i \right) \\ &= \frac{1}{n^2} \sum_{i=1}^{n} \text{var}(z_i) \\ &= \frac{\sigma_z^2}{n} \end{aligned}$$

Therefore, in PPS with replacement, an unbiased estimator for $\bar{Y}$ is

$$\bar{z} = \frac{1}{n} \sum_{i=1}^{n} z_i = \frac{1}{n} \sum_{i=1}^{n} \frac{y_i}{NP_i}$$

with variance given by:

$$\text{var}(\bar{z}) = \frac{\sigma_z^2}{n} = \frac{1}{n} \sum_{i=1}^{N} P_i \left(\frac{Y_i}{NP_i} - \bar{Y} \right)^2$$

If $Y_i = kP_i$ (i.e., Y_i is exactly proportional to P_i), then clearly $\text{var}(\bar{z}) = 0$.

6 The Concedpt of Inclusion Probabilities

Consider a population with units $u_1, u_2, \ldots, u_N$, suppose you want to select a sample s of size n_s . Define the indicator variable:

$$I_i = \begin{cases} 1 & \text{if unit } i \text{ is in the sample} \\ 0 & \text{otherwise} \end{cases}$$

Let π_i be the probability of inclusion of the ith unit of the population in the sample. Then π_i is called the 1st order inclusion probability.

Let π_{ij} be the probability of including both the ith and j^{th} unit of the population in the sample. π_{ij} is called the 2nd order inclusion probability.

Note:

$$E(I_i) = 1 \cdot \Pr(I_i = 1) + 0 \cdot \Pr(I_i = 0) = \Pr(I_i = 1) = \pi_i$$

Next, define

$$n_s = \sum_{i=1}^{N} I_i$$

Therefore

$$E(n_s) = \sum_{i=1}^{N} E(I_i) = \sum_{i=1}^{N} \pi_i$$

If the sample size n_s is fixed, to n i.e. $n_s = n$ then $\sum_{i=1}^{N} \pi_i = n$

$$\text{var}(I_i) = E(I_i^2) - [E(I_i)]^2 = \pi_i - \pi_i^2 = \pi_i(1 - \pi_i)$$

$$E(I_i I_j) = 1 \cdot \Pr(I_i = 1, I_j = 1) = \pi_{ij}$$

Also

$$\begin{aligned}
\sum_{j=1, j\neq i}^{N} \pi_{ij} &= \sum_{j=1, j\neq i}^{N} E(I_i I_j) \\
&= E\left[I_i \sum_{j\neq i}^{N} I_j\right] \\
&= E\left[I_i \left\{\sum_{j=1}^{N} I_j - I_i\right\}\right] \\
&= E\left[I_i\{n - I_i\}\right] \\
&= nE(I_i) - E(I_i^2) \\
&= n\pi_i - \pi_i \\
&= (n-1)\pi_i
\end{aligned}$$

6.1 A Study of Horwitz -Thompson (HT) Estimator)

The HT-estimator for the population mean $\bar{Y}$ is defined as

$$\hat{\bar{Y}}_{HT} = \sum_{i=1}^{n} \frac{y_i}{N\pi_i}$$

where π_i is the probability of including the ith unit from the population into the sample.

Note: Using the indicator variable I_i

$$\hat{\bar{Y}}_{HT} = \sum_{i=1}^{N} \frac{Y_i}{N\pi_i} I_i$$

$$\begin{aligned}
E\left(\hat{\bar{Y}}_{HT}\right) &= \sum_{i=1}^{N} \frac{Y_i}{N\pi_i} E\left(I_i\right) \\
&= \frac{1}{N} \sum_{i=1}^{N} \frac{Y_i}{\pi_i} \pi_i \\
&= \bar{Y}
\end{aligned}$$

This implies that the HT-estimator is an unbiased estimator of $\bar{Y}$.

6.2 Variance of HT- Estimator

$$\text{var}\left(\hat{\bar{Y}}_{HT}\right) = \text{var}\left(\sum_{i=1}^{N} \frac{Y_i}{N\pi_i} I_i\right)$$

$$= \sum_{i=1}^{N} \frac{Y_i^2}{N^2\pi_i^2} \text{var}(I_i) + \sum_{i\neq j}\sum_{j=1}^{N} \frac{Y_iY_j}{N^2\pi_i\pi_j} \text{cov}(I_i, I_j)$$

Since $\text{var}(I_i) = \pi_i(1-\pi_i)$ and $\text{cov}(I_i, I_j) = \pi_{ij} - \pi_i\pi_j$, we have:

$$\text{var}\left(\hat{\bar{Y}}_{HT}\right) = \frac{1}{N^2}\left[\sum_{i=1}^{N} \frac{Y_i^2}{\pi_i^2}\pi_i(1-\pi_i) + \sum_{i\neq j}\sum_{j=1}^{N} \frac{Y_iY_j}{\pi_i\pi_j}(\pi_{ij} - \pi_i\pi_j)\right]$$

$$= \frac{1}{2N^2}\sum_{i=1}^{N}\sum_{j\neq i}(\pi_i\pi_j - \pi_{ij})\left(\frac{Y_i}{\pi_i} - \frac{Y_j}{\pi_j}\right)^2$$

6.3 Variance of the HT - Estimator

An unbiased estimator of $\text{var}\left(\hat{\bar{Y}}_{HT}\right)$ is given by:

$$\hat{V}_{HT}\left(\hat{\bar{Y}}_{HT}\right) = \frac{1}{2N^2}\sum_{i=1}^{n}\sum_{j\neq i}^{n}\left(\frac{\pi_i\pi_j - \pi_{ij}}{\pi_{ij}}\right)\left(\frac{y_i}{\pi_i} - \frac{y_j}{\pi_j}\right)^2$$

Another unbiased estimator of $\text{var}\left(\hat{\bar{Y}}_{HT}\right)$ was given by Yates & Grundy (1953):

$$V_{YG} = \frac{1}{2N^2}\sum_{i=1}^{n}\sum_{j\neq i}^{n}\left(\frac{\pi_i\pi_j - \pi_{ij}}{\pi_{ij}}\right)\left(\frac{y_i}{\pi_i} - \frac{y_j}{\pi_j}\right)^2$$

Advantage: This estimator vanishes when $\pi_i \propto y_i$.

Disadvantage: V_{YG} and $\hat{V}_{HT}$ have a common drawback, which is that they can take negative values since variances should always be positive.

6.4 Relation of the HT - Estimator to the Ratio Estimator under Superpopulation Model

Consider the superpopulation model: $E(Y_i) = \beta x_i$, $\text{var}(Y_i) = \sigma^2 x_i$, $\text{cov}(Y_i, Y_j) = 0$ for $i \neq j$.

Under this model, the HT estimator with $\pi_i \propto x_i$ becomes:

$$\hat{\bar{Y}}_{HT} = \frac{1}{n}\sum_{i=1}^{n}\frac{y_i}{x_i}\bar{X}$$

which is the ratio estimator.

Next;

$$\bar{Y} = \frac{1}{N}\sum_{i=1}^{N} Y_i$$

$$E(\bar{Y}) = \frac{1}{N}\sum_{i=1}^{N} E(Y_i)$$

$$= \frac{1}{N}\sum_{i=1}^{N}\beta x_i$$
$$= \beta\bar{X}$$

Therefore, the HT estimator is model unbiased under the above prediction model.
Next;

$$\hat{\bar{Y}}_{HT} - \bar{Y} = \frac{\bar{X}}{n}\sum_{i=1}^{n}\left(\frac{y_i}{x_i}\right) - \frac{\left(\sum_{i\in s} y_i + \sum_{i\in r} y_i\right)}{N}$$
$$= \frac{\bar{X}}{n}\sum_{i=1}^{n}\left(\frac{y_i}{x_i}\right) - \frac{\sum_{i\in s} y_i}{N} - \frac{\sum_{i\in r} y_i}{N}$$
$$\Rightarrow \text{var}\left(\hat{\bar{Y}}_{HT} - \bar{Y}\right) = \frac{\sigma^2\bar{X}}{N}\left(\frac{\bar{X}N}{n} - 2\frac{\sum_{i=1}^{n} x_i^2}{n} + \frac{\sum_{i=1}^{n} x_i^2}{N\bar{X}}\right)$$
$$= \frac{\sigma^2\bar{X}}{N}\left(\frac{\bar{X}}{f} - 2\bar{x} + \frac{\sum_{i=1}^{n} x_i^2}{N\bar{X}}\right)$$

Hence the optimal sample to use is the one with the largest values of x_i's so as to reduce the variance given above.

Suppose now the model was;

$$\text{var}(Y_i) = V_i, \quad E(Y_i) = \beta_0 + \beta_1 x_i + \beta_2 x_i^2, \quad \text{cov}(Y_i, Y_j) = 0, i \neq j.$$

In this case;

$$E\left(\hat{\bar{Y}}_{HT}\right) = \frac{\bar{X}}{n}\sum_{i=1}^{n} E\left(\frac{\beta_0 + \beta_1 x_i + \beta_2 x_i^2}{x_i}\right)$$
$$= \frac{\beta_0\bar{X}}{n}\sum_{i=1}^{n}\left(\frac{1}{x_i}\right) + \beta_1\bar{X} + \beta_2\bar{X}\bar{x}$$

but

$$E(\bar{Y}) = \beta_0 + \beta_1\bar{X} + \beta_2\frac{\sum_{i=1}^{N} x_i^2}{N}$$

Hence the

$$\text{Bias}\left(\hat{\bar{Y}}_{HT}\right) = \beta_0\left(\frac{\bar{X}}{n}\sum_{i=1}^{n}\left(\frac{1}{x_i}\right) - 1\right) + \beta_2\left(\bar{X}\bar{x} - \frac{\sum_{i=1}^{N} x_i^2}{N}\right)$$

We can remove this bias by using a sample in which $\frac{\bar{X}}{n}\sum_{i=1}^{n}\left(\frac{1}{x_i}\right) = 1$ and $\bar{X}\bar{x} = \frac{\sum_{i=1}^{N} x_i^2}{N}$ such a sample is called a π balanced sample.

6.5 Some Sampling Methods

We now look at some few sampling methods with unequal probability of selection. One of the methods we look into was suggested by Yates and Grundy. In this case, the sample size is 2 i.e $n = 2$. We select the first unit out of N units with PPS. I.e $p_i = \frac{x_i}{X}$ where $X = \sum_{i=1}^{N} x_i$. x_i being the size of the ith unit. We select the second unit out of $N-1$ units with PPS of the remaining units, i.e $p_j = \frac{x_j}{X - x_i} = \frac{p_j}{1-p_i}$.

From this sampling scheme, the inclusion probabilities of unit i and the joint i, j^{th} units is as follows.

The first unit is selected with unequal probabilities of selection. At all subsequent draws, the units are selected with equal probability without replacement. Let π_i be the probability of selection of the first draw for the ith unit. π_i is the Pr(i^{th} unit is selected at the 1st draw) + Pr(i^{th} unit is not selected at the 1st draw but selected in subsequent draws)

$$\begin{aligned}\pi_i &= p_i + (1-p_i)\frac{n-1}{N-1} \\ &= p_i\left(1-\frac{n-1}{N-1}\right)+\frac{n-1}{N-1} \\ &= \left(\frac{N-n}{N-1}\right)p_i+\frac{n-1}{N-1}\end{aligned}$$

π_{ij} is Pr(i^{th} unit is selected at 1st draw; j^{th} unit is selected at any of the $(n-1)$ subsequent draws) + Pr(j^{th} unit is selected at the first draw; i^{th} unit is selected in at any of the $(n-1)$ subsequent draws)

Disadvantage: This sampling procedure is not satisfactory because π_i is not proportional to p_i.

There are many methods described for selection of sample, most of which become complex as n increases. We consider the case when $n=2$ and special cases when $n \geq 2$.

6.5.1 Brewer's Method

Brewer (1963) suggested the following method of selecting a sample of size $n=2$. The first unit is selected with the probability proportional to $\frac{z_i(1-z_i)}{1-2z_i}$, with z_i being a weighted measure of size and $\sum_{i=1}^{N} z_i = 1$. The second unit is selected with probability $\frac{z_j}{1-z_i}$ where i is the unit drawn first. The divisor required to convert $\frac{z_i(1-z_i)}{1-2z_i}$ into actual probabilities is their sum given by:

$$\Delta = \sum_{i=1}^{N}\frac{z_i(1-z_i)}{1-2z_i} = \frac{1}{2}\left[1+\sum_{i=1}^{N}\frac{z_i}{1-2z_i}\right]$$

With $n=2$, the probability that the ith unit is drawn is the sum of the probabilities of the two possible orders:

$$\begin{aligned}\pi_i &= p(i) + \sum_{j(\neq i)}^{N} p(j)p(i/j) \\ &= z_i + \frac{\sum_{j\neq i} z_j z_i}{2\Delta}\left(\frac{1}{1-2z_i}+\frac{1}{1-2z_j}\right) \\ &= 2z_i\end{aligned}$$

This scheme also has $\pi_i\pi_j - \pi_{ij} > 0$ for the Yates-Grundy estimator. Sampford (1967) extended the method to $n>2$ provided $nz_i < 1$ for all units in the population.

7 Systematic Sampling

Objective: To estimate the population mean $\bar{Y}$ by taking a systematic sample of size n out of a population of size N .

Description of a systematic sample: Assume $N = nk$ i.e N is divisible into n groups of size k . Let the units be labeled $1, 2, \ldots, N$ in some order. The corresponding observations be ordered as:

Group	Observations			
1	y_1	y_2	$\cdots$	y_k
2	y_{k+1}	y_{k+2}	$\cdots$	y_{2k}
3	y_{2k+1}	y_{2k+2}	$\cdots$	y_{3k}
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
n	$y_{(n-1)k+1}$	$y_{(n-1)k+2}$	$\cdots$	y_{nk}

Next select an item from group 1 at random (i.e each column with probability $\frac{1}{k}$) suppose the selected unit is in column i with the corresponding observation y_{i1} (the first observation in column i). Then the systematic sample consists of the observations $y_i, y_{i+k}, y_{i+2k}, \ldots, y_{i+(n-1)k}$ the units are each at a distance k, this is also called the ith systematic sample. The possible number of systematic samples each of size n is k.

Let $\bar{y}_{(i)}$ be the mean of the ith systematic sample:

$$\bar{y}_{(i)} = \frac{1}{n}\sum_{j=1}^{n} y_{i+(j-1)k}$$

Since each systematic sample has probability $\frac{1}{k}$ of being selected,

$$E(\bar{y}_{sy}) = \frac{1}{k}\sum_{i=1}^{k}\bar{y}_{(i)} = \frac{1}{k}\sum_{i=1}^{k}\frac{1}{n}\sum_{j=1}^{n} y_{i+(j-1)k} = \frac{1}{N}\sum_{i=1}^{N} y_i = \bar{Y}$$

Hence $\bar{y}_{sy}$ is an unbiased estimator of $\bar{Y}$.

7.1 Variance of Systematic Sample Mean

By definition,

$$\text{var}(\bar{y}_{sy}) = E(\bar{y}_{sy} - \bar{Y})^2 = \frac{1}{k}\sum_{i=1}^{k}(\bar{y}_{(i)} - \bar{Y})^2$$

Define the total sum of squares:

$$S^2 = \frac{1}{N-1}\sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij} - \bar{Y})^2$$

We can decompose this into within-sample and between-sample components. It can be shown that:

$$\text{var}(\bar{y}_{sy}) = \frac{S^2}{n}\left[1 + (n-1)\rho_w\right]\left(\frac{N-n}{N}\right)$$

where ρ_w is the intraclass correlation coefficient within the systematic samples:

$$\rho_w = \frac{E(y_{ij} - \bar{Y})(y_{i\ell} - \bar{Y})}{E(y_{ij} - \bar{Y})^2}$$

Alternatively, we can write:

$$\text{var}(\bar{y}_{sy}) = \frac{N-1}{N}S^2 - \frac{k(n-1)}{N}S^2_{wsy}$$

where S^2_{wsy} is the variance among units within the same systematic sample.

7.2 Comparison of Systematic Sampling with SRSWOR

$$\frac{\text{var}(\bar{y}_{sy})}{\text{var}(\bar{y}_{SRSWOR})} = \frac{\frac{S^2}{n}\left[1+(n-1)\rho_w\right]\left(\frac{N-n}{N}\right)}{\frac{S^2}{n}\left(\frac{N-n}{N}\right)} = 1+(n-1)\rho_w$$

Thus:

- If $\rho_w > 0$, systematic sampling is less efficient than SRSWOR.
- If $\rho_w = 0$, systematic sampling is equally efficient as SRSWOR.
- If $\rho_w < 0$, systematic sampling is more efficient than SRSWOR.

Systematic sampling is more efficient than SRSWOR if:

$$1+(n-1)\rho_w < 1 \Rightarrow (n-1)\rho_w < 0 \Rightarrow \rho_w < 0$$

The condition $\rho_w < -\frac{1}{N-1}$ is required for $\text{var}(\bar{y}_{sy}) < \text{var}(\bar{y}_{SRSWOR})$.

7.3 Estimation of Variance in Systematic Sampling

A systematic sample is a SRS of only one cluster. Therefore, no unbiased estimate of variance can be formed from the sample. This is a severe drawback of systematic sampling. A number of approximations for estimation of the variance are available.

One method is to treat the systematic sample as if it were a SRS of n units and estimate the variance by:

$$\hat{\text{var}}(\bar{y}_{sy}) = \left(\frac{1}{n} - \frac{1}{N}\right) s_w^2$$

where $s_w^2 = \frac{1}{n-1}\sum_{j=1}^{n}(y_{ij} - \bar{y}_{(i)})^2$ is the variance within the selected systematic sample.

Observe:

$$E\left[\left(\frac{1}{n} - \frac{1}{N}\right) s_w^2\right] = \left(\frac{1}{n} - \frac{1}{N}\right) E\left(s_w^2\right)$$

$$E\left(s_w^2\right) = \frac{1}{n-1}\left\{E\left(\sum_{j=1}^{n} y_{ij}^2\right) - n\left[E\left(\bar{y}_i^2\right)\right]\right\}$$

Note $\sum_{j=1}^{n} y_{ij}^2$ can take the values $\sum_{j=1}^{n} y_{1j}^2, \sum_{j=1}^{n} y_{2j}^2, \ldots \sum_{j=1}^{n} y_{kj}^2$ all having the probability $\frac{1}{k}$.

7.4 Exercise 15

Compare $E(s_w^2) = \left(\frac{nk-1}{nk}\right) S^2[1-\rho_w]$ with $\text{var}(\bar{y}_{sy}) = \left(\frac{nk-1}{nk}\right) S^2[1-(n-1)\rho_w]$

7.5 Cluster Sampling

A sampling procedure assumes the division of the population into a finite number of distinct and identifiable units called the sampling units. The smallest units into which the population can be divided are called elements of the population and groups of elements are called clusters. When the sampling unit is a cluster, the procedure of sampling is called cluster sampling. When the list of elements (frame) is not available, using an element as the sampling unit is not feasible. The method of cluster sampling can be used in such cases e.g Lists of farms may not be available. A necessary condition for the procedure is that every unit of the population must correspond to one and only one cluster.

7.5.1 Notations for Cluster Sampling with Equal Cluster Sizes

Assume the population consists of N clusters, each of size M. Let:

$$Y_{ij} = \text{value for the } j^{th} \text{ element in the } i^{th} \text{ cluster}$$

Define:

$$Y_i = \sum_{j=1}^{M} Y_{ij} = \text{total for cluster } i$$

$$\bar{Y}_i = \frac{Y_i}{M} = \text{mean for cluster } i$$

$$\bar{Y} = \frac{1}{NM} \sum_{i=1}^{N} \sum_{j=1}^{M} Y_{ij} = \frac{1}{N} \sum_{i=1}^{N} \bar{Y}_i$$

If we select n clusters by SRSWOR and then include all elements in the selected clusters, the estimator of the population mean is:

$$\bar{y}_{cl} = \frac{1}{nM} \sum_{i=1}^{n} \sum_{j=1}^{M} y_{ij} = \frac{1}{n} \sum_{i=1}^{n} \bar{y}_i$$

7.5.2 Properties of the Cluster Sample Mean

Since the clusters are selected by SRSWOR:

$$E(\bar{y}_{cl}) = \frac{1}{n} \sum_{i=1}^{n} E(\bar{y}_i) = \frac{1}{n} \sum_{i=1}^{n} \bar{Y}_i = \bar{Y}$$

Thus $\bar{y}_{cl}$ is unbiased.

The variance is:

$$\text{var}(\bar{y}_{cl}) = \frac{1-f}{n} \cdot \frac{1}{N-1} \sum_{i=1}^{N} (\bar{Y}_i - \bar{Y})^2$$

where $f = \frac{n}{N}$.

Let

$$S_b^2 = \frac{1}{N-1} \sum_{i=1}^{N} (\bar{Y}_i - \bar{Y})^2$$

be the between-cluster variance. Then:

$$\text{var}(\bar{y}_{cl}) = \frac{1-f}{n} S_b^2$$

7.5.3 Comparison of Cluster Sampling with SRSWOR

If we had taken an SRSWOR of nM elements directly, the variance would be:

$$\text{var}(\bar{y}_{SRS}) = \frac{1-f'}{nM} S^2$$

where $f' = \frac{nM}{NM} = \frac{n}{N} = f$ and S^2 is the element variance.

The ratio of the variances is:

$$\frac{\text{var}(\bar{y}_{cl})}{\text{var}(\bar{y}_{SRS})} = \frac{\frac{1-f}{n} S_b^2}{\frac{1-f}{nM} S^2} = M \cdot \frac{S_b^2}{S^2}$$

But S^2 can be decomposed as:

$$S^2 = \frac{M-1}{M} S_w^2 + S_b^2$$

where $S_w^2 = \frac{1}{N(M-1)} \sum_{i=1}^{N} \sum_{j=1}^{M} (Y_{ij} - \bar{Y}_i)^2$ is the within-cluster variance.

Thus:

$$\frac{\text{var}(\bar{y}_{cl})}{\text{var}(\bar{y}_{SRS})} = 1 + \frac{(M-1)S_w^2}{MS_b^2}$$

Cluster sampling is less efficient than SRSWOR unless the clusters are homogeneous (small S_w^2) or the between-cluster variance is large.

7.6 Two Stage Sampling (FSU - Equal Sizes)

We now consider a case when a specified number of elements is selected from each of the selected cluster. Such a procedure is known as sub sampling procedure or two stage sampling because the sample is chosen in two stages. The clusters that form the units of the sampling at the first stage are called first stage units or primary sampling units (PSU or FSU). And the elements or groups of elements within the clusters which form units of sampling at the second stage are called second stage sampling units (SSU). The procedure is generalised to three or more stages and is termed as multistage sampling.

We will assume that sampling at each stage is carried out with equal probability and without replacement, i.e SRSWOR at each stage will be used.

Assume that the population has NM elements, grouped in N clusters each of size M.

Let Y_{ij} be the jth measurement of the ith cluster (survey measurement of the jth unit in the ith cluster).

$$\bar{Y}_i = \frac{1}{M} \sum_{j=1}^{M} Y_{ij} \quad \text{(average of the survey measurements in the } i^{th} \text{ cluster)}$$

$$\bar{Y} = \frac{1}{NM} \sum_{i=1}^{N} \sum_{j=1}^{M} Y_{ij} \quad \text{(population mean)}$$

Next assume that a sample of size nm has been drawn by first selecting n PSU and then selecting m SSU from each of the selected PSU. Let:

$$\bar{y}_i = \frac{1}{m} \sum_{j=1}^{m} y_{ij} \quad \text{(average of elements in the } i^{th} \text{ selected PSU)}$$

Consider

$$\bar{\bar{y}} = \frac{1}{nm} \sum_{i=1}^{n} \sum_{j=1}^{m} y_{ij} = \frac{1}{n} \sum_{i=1}^{n} \bar{y}_i$$

as an estimator of $\bar{Y}$. Is this a good estimator?

Recall the double expectation theorem:

$$E(\hat{\theta}) = E_1[E_2(\hat{\theta})]$$

$$\text{var}(\hat{\theta}) = \text{var}_1[E_2(\hat{\theta})] + E_1[\text{var}_2(\hat{\theta})]$$

Now using this, observe:

$$E(\bar{\bar{y}}) = E_1[E_2(\bar{\bar{y}})]$$

First, conditional on the selected PSU:

$$E_2(\bar{\bar{y}}) = E_2\left(\frac{1}{n}\sum_{i=1}^{n}\bar{y}_i\right) = \frac{1}{n}\sum_{i=1}^{n}E_2(\bar{y}_i)$$

For a given selected PSU i, under SRSWOR of m elements from M:

$$E_2(\bar{y}_i) = \bar{Y}_i$$

$$\text{var}_2(\bar{y}_i) = \frac{M-m}{M}\cdot\frac{S_i^2}{m}$$

where $S_i^2 = \frac{1}{M-1}\sum_{j=1}^{M}(Y_{ij}-\bar{Y}_i)^2$ is the population variance within the ith PSU.

Thus:

$$E_2(\bar{\bar{y}}) = \frac{1}{n}\sum_{i=1}^{n}\bar{Y}_i$$

Now taking expectation over the first stage:

$$E(\bar{\bar{y}}) = E_1\left(\frac{1}{n}\sum_{i=1}^{n}\bar{Y}_i\right) = \frac{1}{N}\sum_{i=1}^{N}\bar{Y}_i = \bar{Y}$$

Hence $\bar{\bar{y}}$ is an unbiased estimator of $\bar{Y}$.

For the variance:

$$\text{var}(\bar{\bar{y}}) = \text{var}_1[E_2(\bar{\bar{y}})] + E_1[\text{var}_2(\bar{\bar{y}})]$$

First term:

$$\text{var}_1[E_2(\bar{\bar{y}})] = \text{var}_1\left(\frac{1}{n}\sum_{i=1}^{n}\bar{Y}_i\right) = \frac{1-f_1}{n}S_b^2$$

where $f_1 = \frac{n}{N}$ and $S_b^2 = \frac{1}{N-1}\sum_{i=1}^{N}(\bar{Y}_i-\bar{Y})^2$ is the between-PSU variance.

Second term:

$$E_1[\text{var}_2(\bar{\bar{y}})] = E_1\left[\text{var}_2\left(\frac{1}{n}\sum_{i=1}^{n}\bar{y}_i\right)\right] = E_1\left[\frac{1}{n^2}\sum_{i=1}^{n}\text{var}_2(\bar{y}_i)\right]$$

$$= E_1\left[\frac{1}{n^2}\sum_{i=1}^{n}\frac{M-m}{M}\cdot\frac{S_i^2}{m}\right] = \frac{1-f_2}{nm}E_1\left[\frac{1}{n}\sum_{i=1}^{n}S_i^2\right]$$

where $f_2 = \frac{m}{M}$.

But $E_1\left[\frac{1}{n}\sum_{i=1}^{n}S_i^2\right] = \frac{1}{N}\sum_{i=1}^{N}S_i^2 = \bar{S}_w^2$, the average within-PSU variance.

Thus:

$$\text{var}(\bar{\bar{y}}) = \frac{1-f_1}{n}S_b^2 + \frac{1-f_2}{nm}\bar{S}_w^2$$

7.7 Two Stage Sampling : Unequal FSU

We now consider the case where FSU are of unequal sizes and SRSWOR is employed at each stage.

Notations:

$$M_i = \text{Number of SSU in the } i^{th} \text{ PSU}$$

$$\bar{M} = \frac{1}{N}\sum_{i=1}^{N}M_i$$

$$\bar{Y}_i = \frac{1}{M_i}\sum_{j=1}^{M_i} Y_{ij}$$

$$\bar{Y} = \frac{1}{N\bar{M}}\sum_{i=1}^{N} M_i\bar{Y}_i = \frac{1}{\sum M_i}\sum_{i=1}^{N} M_i\bar{Y}_i$$

$$S_{iy}^2 = \frac{1}{M_i - 1}\sum_{j=1}^{M_i}(Y_{ij} - \bar{Y}_i)^2$$

$$S_{by}^2 = \frac{1}{N-1}\sum_{i=1}^{N}\left(\frac{M_i}{\bar{M}}\bar{Y}_i - \bar{Y}\right)^2$$

$$m_i = \text{Number of SSU selected from the } i^{th} \text{ sample PSU}$$

$$\bar{y}_i = \frac{1}{m_i}\sum_{j=1}^{m_i} y_{ij}$$

Our interest is to estimate population mean $\bar{Y}$. There exist a number of sampling strategies for choosing this.

7.7.1 Estimator 1: Simple Mean of PSU Means

$$\hat{\bar{Y}}_1 = \frac{1}{n}\sum_{i=1}^{n}\bar{y}_i$$

Is $\hat{\bar{Y}}_1$ unbiased?

$$E(\hat{\bar{Y}}_1) = E_1E_2\left(\frac{1}{n}\sum_{i=1}^{n}\bar{y}_i\right) = E_1\left(\frac{1}{n}\sum_{i=1}^{n}\bar{Y}_i\right) = \frac{1}{N}\sum_{i=1}^{N}\bar{Y}_i$$

But $\bar{Y} = \frac{1}{N\bar{M}}\sum_{i=1}^{N} M_i\bar{Y}_i$. Unless all M_i are equal, $\frac{1}{N}\sum\bar{Y}_i \neq \bar{Y}$. Thus $\hat{\bar{Y}}_1$ is biased.

7.7.2 Estimator 2: Weighted Mean with PSU Sizes

$$\hat{\bar{Y}}_2 = \frac{1}{n\bar{M}}\sum_{i=1}^{n} M_i\bar{y}_i$$

$$E(\hat{\bar{Y}}_2) = E_1E_2\left(\frac{1}{n\bar{M}}\sum_{i=1}^{n} M_i\bar{y}_i\right) = E_1\left(\frac{1}{n\bar{M}}\sum_{i=1}^{n} M_i\bar{Y}_i\right)$$

$$= \frac{1}{N\bar{M}}\sum_{i=1}^{N} M_i\bar{Y}_i = \bar{Y}$$

Thus $\hat{\bar{Y}}_2$ is unbiased.

7.7.3 Variance of $\hat{\bar{Y}}_2$

$$\text{var}(\hat{\bar{Y}}_2) = \text{var}_1[E_2(\hat{\bar{Y}}_2)] + E_1[\text{var}_2(\hat{\bar{Y}}_2)]$$

First term:

$$\text{var}_1[E_2(\hat{\bar{Y}}_2)] = \text{var}_1\left(\frac{1}{n\bar{M}}\sum_{i=1}^{n} M_i\bar{Y}_i\right) = \frac{1-f_1}{n}S_u^2$$

where

$$S_u^2 = \frac{1}{N-1}\sum_{i=1}^{N}\left(\frac{M_i}{\bar{M}}\bar{Y}_i - \bar{Y}\right)^2$$

Second term:

$$E_1[\text{var}_2(\hat{\bar{Y}}_2)] = E_1\left[\text{var}_2\left(\frac{1}{n\bar{M}}\sum_{i=1}^{n} M_i\bar{y}_i\right)\right] = E_1\left[\frac{1}{n^2\bar{M}^2}\sum_{i=1}^{n} M_i^2\,\text{var}_2(\bar{y}_i)\right]$$

For a given selected PSU i:

$$\text{var}_2(\bar{y}_i) = \frac{M_i - m_i}{M_i}\cdot\frac{S_{iy}^2}{m_i}$$

Thus:

$$E_1[\text{var}_2(\hat{\bar{Y}}_2)] = E_1\left[\frac{1}{n^2\bar{M}^2}\sum_{i=1}^{n} M_i^2\cdot\frac{M_i - m_i}{M_i}\cdot\frac{S_{iy}^2}{m_i}\right]$$

$$= \frac{1}{n\bar{M}^2}\cdot\frac{1}{N}\sum_{i=1}^{N} M_i(M_i - m_i)\frac{S_{iy}^2}{m_i}$$

Therefore:

$$\text{var}(\hat{\bar{Y}}_2) = \frac{1-f_1}{n}S_u^2 + \frac{1}{nN\bar{M}^2}\sum_{i=1}^{N} M_i(M_i - m_i)\frac{S_{iy}^2}{m_i}$$

7.7.4 Example

A videotape hire company has shops in each of the 5 regions: three regions have 12 shops, the others just have 8. To estimate the total number Y_T of video films hired from the company in a particular week, the sales manager phones 12 shops chosen by picking 3 regions at random then making a choice of 3,5 and 4 shops from the chosen regions. The results were as follows:

First region: 260, 296, 182 - 12 shops Second region: 156, 261, 130, 302, 241 - 8 shops Third region: 196, 356, 268, 284 - 12 shops

Solution:

The two estimates of Y_T are given by:

$$\hat{Y}_T = \frac{M}{m}\sum_{i=1}^{m} L_i\bar{y}_i$$

where $L_i = 12, 12, 12, 8, 8$, $N = 5$, $M = 5$, and $m = 3$ (number of regions sampled).

$$\bar{y}_1 = \frac{260 + 296 + 182}{3} = \frac{738}{3} = 246$$

$$\bar{y}_2 = \frac{156 + 261 + 130 + 302 + 241}{5} = \frac{1090}{5} = 218$$

$$\bar{y}_3 = \frac{196 + 356 + 268 + 284}{4} = \frac{1104}{4} = 276$$

$$s_1^2 = \frac{(260-246)^2 + (296-246)^2 + (182-246)^2}{2} = \frac{196 + 2500 + 4096}{2} = \frac{6792}{2} = 3396$$

$$s_2^2 = \frac{(156-218)^2 + (261-218)^2 + (130-218)^2 + (302-218)^2 + (241-218)^2}{4}$$

$$= \frac{3844 + 1849 + 7744 + 7056 + 529}{4} = \frac{21022}{4} = 5255.5$$

$$s_3^2 = \frac{(196-276)^2 + (356-276)^2 + (268-276)^2 + (284-276)^2}{3}$$

$$= \frac{6400 + 6400 + 64 + 64}{3} = \frac{12928}{3} = 4309.33$$

Now, using estimator $\hat{\bar{Y}}_2$:

$$\hat{\bar{Y}}_2 = \frac{1}{n\bar{M}} \sum_{i=1}^{n} M_i \bar{y}_i$$

where $n = 3$, $\bar{M} = \frac{12+12+12+8+8}{5} = \frac{52}{5} = 10.4$

$$\sum_{i=1}^{3} M_i \bar{y}_i = 12 \times 246 + 8 \times 218 + 12 \times 276 = 2952 + 1744 + 3312 = 8008$$

$$\hat{\bar{Y}}_2 = \frac{8008}{3 \times 10.4} = \frac{8008}{31.2} = 256.67$$

$$\hat{Y}_T = N\bar{M}\hat{\bar{Y}}_2 = 5 \times 10.4 \times 256.67 = 52 \times 256.67 = 13346.84$$

7.7.5 Exercise

In two stage sampling with unequal first sampling units, let $\hat{\bar{y}} = \frac{1}{n}\sum_{i=1}^{n} \bar{y}_i$ be an estimator of the population mean $\bar{Y}$ where $\bar{y}_i = \frac{1}{m_i}\sum_{j=1}^{m_i} y_{ij}$ with y_{ij} being the jth unit in the ith cluster. Test $\hat{\bar{y}}$ for unbiasedness. Suppose that we now define another estimator, $\hat{\bar{y}}_1 = \frac{1}{n}\sum_{i=1}^{n} \frac{M_i}{\bar{M}} \bar{y}_i$ with $\frac{M_i}{\bar{M}}$ being the weight of the ith stratum. Compare $\hat{\bar{y}}$ to $\hat{\bar{y}}_1$ and choose the best estimator of the population mean $\bar{Y}$ using unbiasedness criteria.

Solution

For $\hat{\bar{y}} = \frac{1}{n}\sum_{i=1}^{n} \bar{y}_i$:

$$E(\hat{\bar{y}}) = E_1 E_2 \left(\frac{1}{n} \sum_{i=1}^{n} \bar{y}_i \right) = E_1 \left(\frac{1}{n} \sum_{i=1}^{n} E_2(\bar{y}_i) \right)$$

$$= E_1 \left(\frac{1}{n} \sum_{i=1}^{n} \bar{Y}_i \right) = \frac{1}{N} \sum_{i=1}^{N} \bar{Y}_i$$

But $\bar{Y} = \frac{1}{N\bar{M}}\sum_{i=1}^{N} M_i \bar{Y}_i$. Unless all M_i are equal, $\frac{1}{N}\sum \bar{Y}_i \neq \bar{Y}$. Hence $\hat{\bar{y}}$ is biased.
For $\hat{\bar{y}}_1 = \frac{1}{n}\sum_{i=1}^{n} \frac{M_i}{\bar{M}} \bar{y}_i$:

$$E(\hat{\bar{y}}_1) = E_1 E_2 \left(\frac{1}{n} \sum_{i=1}^{n} \frac{M_i}{\bar{M}} \bar{y}_i \right) = E_1 \left(\frac{1}{n} \sum_{i=1}^{n} \frac{M_i}{\bar{M}} \bar{Y}_i \right)$$

$$= \frac{1}{N}\sum_{i=1}^{N}\frac{M_i}{\bar{M}}\bar{Y}_i = \frac{1}{N\bar{M}}\sum_{i=1}^{N}M_i\bar{Y}_i = \bar{Y}$$

Hence $\hat{\bar{y}}_1$ is unbiased. Using unbiasedness criteria, $\hat{\bar{y}}_1$ is the better estimator of the population mean $\bar{Y}$ compared to $\hat{\bar{y}}$.

7.7.6 Exercise

A company, which provides its salesmen with cars for company business only, wishes to estimate the average number of miles covered by each car last year. The company operates from 12 branches, and the numbers of cars, N_i and means and variances ($\bar{Y}_i$ and S_i^2) of miles driven last year (in thousands of miles) for each branch, are as follows.

Branch	N_i	$\bar{Y}_i$	S_i^2
1	6	24.32	5.07
2	7	27.06	5.53
3	11	27.60	6.24
4	7	28.01	6.59
5	8	27.56	6.21
6	14	29.07	6.12
7	6	32.03	5.97
8	2	28.41	6.01
9	2	28.91	5.74
10	5	25.55	6.78
11	12	28.58	5.87
12	6	27.27	5.38

Suppose that the average mileage is to be estimated by sampling a few branches at random, and using the figures for all the cars at the chosen branches. Work out the standard errors of the unbiased cluster sample estimator $\bar{y}_{cl}$ for a cluster of 4 branches, and a simple random sample of 27 cars (this being approximately the average numbers of cars which would be obtained in a cluster of 4 branches). Compare the efficiencies of the two methods.

8 Double Sampling

This is a technique in which we select, say n' units in the first phase from N units (1st phase) and select a sub sample of n units from the n' units (2nd phase). The method is very useful in ratio and regression estimation, where an auxiliary variate x_i is required but is unknown. The first phase in the sampling helps to estimate x (auxiliary variable) and the second phase helps to estimate y (the study variable).

Let $\hat{\theta}$ be the estimate of θ from double sampling. Then

$$E\left(\hat{\theta}\right) = E_1\left[E_2\left(\hat{\theta}\right)\right]$$

where E_1 is expectation over the first phase sampling and E_2 is expectation over the second phase sampling given the first phase sample.

$$\text{var}\left(\hat{\theta}\right) = \text{var}_1\left[E_2\left(\hat{\theta}\right)\right] + E_1\left[\text{var}_2\left(\hat{\theta}\right)\right]$$

8.1 Double Sampling for Ratio Estimation

Suppose we want to estimate the population mean $\bar{Y}$ using ratio estimation, but the population mean $\bar{X}$ of the auxiliary variable is unknown. We take a large first-phase sample of size n' to estimate $\bar{X}$, and a smaller second-phase sample of size n to measure both y and x.

Let:

- From the first-phase sample (size n'): measure only x, compute $\bar{x}'$
- From the second-phase sample (size $n < n'$): measure both y and x, compute $\bar{y}$, $\bar{x}$, and $\hat{R} = \bar{y}/\bar{x}$

The double sampling ratio estimator is:

$$\bar{y}_{Rd} = \hat{R}\bar{x}' = \frac{\bar{y}}{\bar{x}}\bar{x}'$$

8.1.1 Approximate Variance of $\bar{y}_{Rd}$

To the first order of approximation:

$$\text{var}(\bar{y}_{Rd}) \approx \frac{1-f'}{n'}S_y^2 + \frac{1-f}{n}(S_y^2 - 2RS_{xy} + R^2S_x^2)$$

where $f' = n'/N$ and $f = n/n'$ (the second-phase sampling fraction relative to the first-phase sample).

Alternatively:

$$\text{var}(\bar{y}_{Rd}) \approx \frac{1}{n}S_y^2 + \left(\frac{1}{n'} - \frac{1}{n}\right)(S_y^2 - 2RS_{xy} + R^2S_x^2)$$

8.2 Double Sampling for Post-Stratification

In this case, all the units in the population are considered without stratification. The reason being the stratum sizes N_h are unknown. By simple random sampling, we select n' units. We then categorise the selected units into H strata where n'_h are units from n' falling into stratum h. Let

$$W_h = \frac{N_h}{N} \quad \text{be population proportion in stratum } h.$$

$$w_h = \frac{n'_h}{n'} \quad \text{is proportion of first-phase sample units in stratum } h.$$

In the second phase, we select n_h units from the n'_h units in stratum h. The objective of the first sample is to estimate the strata weights W_h while that of the second phase is to estimate the strata means $\bar{Y}_h$.

The population mean

$$\bar{Y} = \frac{1}{N}\sum_{h=1}^{H} N_h\bar{Y}_h = \sum_{h=1}^{H} W_h\bar{Y}_h$$

This is estimated by:

$$\bar{y}_{pstd} = \sum_{h=1}^{H} w_h\bar{y}_h$$

where $\bar{y}_h = \frac{1}{n_h}\sum_{i=1}^{n_h} y_{hi}$ is the sample mean from stratum h in the second phase.

8.2.1 Properties of $\bar{y}_{pstd}$

$$E(\bar{y}_{pstd}) = E_1\left[\sum_{h=1}^{H} w_h E_2(\bar{y}_h|\text{first phase})\right] = E_1\left[\sum_{h=1}^{H} w_h \bar{Y}_h\right]$$

Since $E_1(w_h) = W_h$ and $\bar{Y}_h$ is constant,

$$E(\bar{y}_{pstd}) = \sum_{h=1}^{H} W_h \bar{Y}_h = \bar{Y}$$

Thus $\bar{y}_{pstd}$ is unbiased.
For the variance:

$$\text{var}(\bar{y}_{pstd}) = \text{var}_1[E_2(\bar{y}_{pstd})] + E_1[\text{var}_2(\bar{y}_{pstd})]$$

First term:

$$\text{var}_1[E_2(\bar{y}_{pstd})] = \text{var}_1\left(\sum_{h=1}^{H} w_h \bar{Y}_h\right) = \sum_{h=1}^{H} \bar{Y}_h^2 \,\text{var}_1(w_h) + \sum_{h\neq k} \bar{Y}_h \bar{Y}_k \,\text{cov}_1(w_h, w_k)$$

Since $(n'_1, \ldots, n'_H)$ follows a multinomial distribution with parameters n' and $(W_1, \ldots, W_H)$, we have:

$$\text{var}_1(w_h) = \frac{W_h(1 - W_h)}{n'}, \quad \text{cov}_1(w_h, w_k) = -\frac{W_h W_k}{n'}$$

Thus:

$$\text{var}_1[E_2(\bar{y}_{pstd})] = \frac{1}{n'}\left[\sum_{h=1}^{H} W_h \bar{Y}_h^2 - \left(\sum_{h=1}^{H} W_h \bar{Y}_h\right)^2\right] = \frac{1}{n'} S_b^2$$

where $S_b^2 = \sum_{h=1}^{H} W_h(\bar{Y}_h - \bar{Y})^2$ is the between-stratum variance.
Second term:

$$E_1[\text{var}_2(\bar{y}_{pstd})] = E_1\left[\sum_{h=1}^{H} w_h^2 \,\text{var}_2(\bar{y}_h)\right]$$

Given the first-phase sample, the second-phase sample in stratum h is a SRS of size n_h from n'_h units. If we assume the second-phase sampling fraction is small, $\text{var}_2(\bar{y}_h) \approx S_h^2/n_h$. Then:

$$E_1[\text{var}_2(\bar{y}_{pstd})] \approx E_1\left[\sum_{h=1}^{H} \frac{w_h^2 S_h^2}{n_h}\right]$$

For a fixed second-phase sample size $n = \sum n_h$, and with proportional allocation $n_h = n w_h$:

$$E_1[\text{var}_2(\bar{y}_{pstd})] \approx E_1\left[\sum_{h=1}^{H} \frac{w_h S_h^2}{n}\right] = \frac{1}{n}\sum_{h=1}^{H} W_h S_h^2$$

Thus:

$$\text{var}(\bar{y}_{pstd}) \approx \frac{1}{n'} S_b^2 + \frac{1}{n}\bar{S}_w^2$$

where $\bar{S}_w^2 = \sum_{h=1}^{H} W_h S_h^2$ is the average within-stratum variance.

9 Sampling Over Time (Repeated Surveys)

There are three types of repeated surveys.

9.1 Panel (Longitudinal Surveys)

$$\text{at } t_1 : y_{11}, y_{12}, \ldots, y_{1n} \quad \text{and} \quad \text{at } t_2 : y_{21}, y_{22}, \ldots, y_{2n}$$

same individuals are measured at both times.

9.2 Repeated Cross-Sectional Survey

At $t_1 : y_{11}, y_{12}, \ldots, y_{1n}$, and at t_2 select an independent sample of size n from the population then you get the measurements $y_{21}, y_{22}, \ldots, y_{2n}$.

9.3 Rotational Survey Designs (Partially Overlapping Design)

At time t_2, we have:

$$\underbrace{y_{21}, y_{22}, \ldots, y_{2m}}_{m \text{ from previous sample}} \quad \underbrace{y_{2,m+1}, y_{2,m+2}, \ldots, y_{2n}}_{n-m \text{ new units}}$$

Looking at the above three designs, we note that:

- If $m = n$, then design three is the same as design one.
- If $m = 0$, then design three is the same as design two.

If therefore we can analyse design three, the rest can be treated as special cases.

9.4 Estimation of Change

Let $\bar{y}_1$ be the sample mean at time t_1 based on n units. Let $\bar{y}_2$ be the sample mean at time t_2 based on n units, with m units matched from time t_1.

The estimator of change $\Delta = \bar{Y}_2 - \bar{Y}_1$ is $\hat{\Delta} = \bar{y}_2 - \bar{y}_1$.

$$\text{var}(\hat{\Delta}) = \text{var}(\bar{y}_1) + \text{var}(\bar{y}_2) - 2\,\text{cov}(\bar{y}_1, \bar{y}_2)$$

Assuming the population variances are equal at both times ($S_1^2 = S_2^2 = S^2$):

$$\text{var}(\bar{y}_1) = \text{var}(\bar{y}_2) = \frac{1-f}{n} S^2$$

where $f = n/N$.

The covariance term depends on the matched portion. If the matched units are a subsample of the first sample, and assuming the correlation between the same unit's values at two times is ρ, then:

$$\text{cov}(\bar{y}_1, \bar{y}_2) = \frac{m}{n^2} \rho S^2$$

Thus:

$$\text{var}(\hat{\Delta}) = 2\frac{1-f}{n} S^2 - 2\frac{m}{n^2} \rho S^2 = 2S^2 \left(\frac{1}{n} - \frac{1}{N} \right) \left(1 - \frac{m\rho}{n} \right)$$

For fixed total sample size, the variance is minimized when m is as large as possible (i.e., $m = n$) if $\rho > 0$.

9.5 Estimation of Current Mean at Time t_2

We can use a composite estimator combining the matched and unmatched portions. There are two ways to select the unmatched portion:

1. Whether the $(n-m)$ units were selected from the remaining $N-m$ units after selecting m matched units, or
2. Whether they were selected from all units independently.

If case 2 is true, then $\bar{y}_{2m}$ and $\bar{y}_{2u}$ will be independent.

$$\text{var}(\bar{y}_2) = \left(\frac{m}{n}\right)^2 \text{var}(\bar{y}_{2m}) + \left(\frac{n-m}{n}\right)^2 \text{var}(\bar{y}_{2u})$$

$$= \frac{m^2}{n^2} \cdot \frac{S^2}{m}\left(1 - \frac{m}{N}\right) + \left(\frac{n-m}{n}\right)^2 \left(1 - \frac{n-m}{N}\right) \frac{S^2}{n-m}$$

$$= \frac{S^2}{n}\left[1 - \frac{m^2 + (n-m)^2}{nN}\right]$$

Now, for a regression estimator using the matched portion:

$$\bar{y}_{2r} = \bar{y}_{2m} + b(\bar{y}_1 - \bar{y}_{1m})$$

where b is the sample regression coefficient from the matched units.

$$\text{var}(\bar{y}_{2r}) \approx \frac{S^2}{n}\left[1 - \frac{m^2}{nN} - \frac{(n-m)^2}{nN}\right]?$$

More precisely, we need the covariance between $\bar{y}_1$ and $\bar{y}_2$:

$$\text{cov}(\bar{y}_1, \bar{y}_2) = \text{cov}_1(E_2(\bar{y}_1), E_2(\bar{y}_2)) + E_1[\text{cov}_2(\bar{y}_1, \bar{y}_2)]$$

We note that $E_2(\bar{y}_1) = \bar{y}_1$ because $\bar{y}_1$ is fixed at $t = 2$. And $E_2(\bar{y}_2) = \frac{1}{n}\sum_{i=1}^{n} y_{2i}$.
$\text{cov}_2[\bar{y}_1, \bar{y}_2] = 0$ since $\bar{y}_1$ on occasion two is fixed.
Thus:

$$\text{cov}(\bar{y}_1, \bar{y}_2) = \text{cov}_1(\bar{y}_1, \bar{y}_2')$$

9.5.1 Optimal Estimation of $\bar{Y}_2$

Treat the matched sample as a double sample i.e in first phase matched SRS of size n and in phase two, we take a simple random sub-sample of size m from n units. Using the regression estimator,

$$\bar{y}_{2r} = \bar{y}_{2m} + b(\bar{y}_1 - \bar{y}_{1m})$$

where $\bar{y}_{1m}$ is the mean of the matched units at $t = 1$ and $\bar{y}_{2m}$ is the mean of matched units at $t = 2$. b is computed from the m matched units.

$$\hat{b} = \frac{\sum_{i=1}^{m}(y_{1i} - \bar{y}_{1m})(y_{2i} - \bar{y}_{2m})}{\sum_{i=1}^{m}(y_{1i} - \bar{y}_{1m})^2}$$

We can now estimate $\bar{y}_2$ by using the unmatched units and the matched units via a composite estimator.

9.5.2 Composite Estimator

$$\bar{y}_{com} = \phi\bar{y}_{2u} + (1-\phi)\bar{y}_{2r}$$

$$\text{var}(\bar{y}_{com}) = \phi^2\,\text{var}(\bar{y}_{2u}) + (1-\phi)^2\,\text{var}(\bar{y}_{2r})$$

Because $\bar{y}_{2u}$ and $\bar{y}_{2r}$ are independent (under case 2), The optimal value of ϕ is obtained from:

$$\frac{\partial}{\partial\phi}\,\text{var}(\bar{y}_{com}) = 2\phi\,\text{var}(\bar{y}_{2u}) - 2(1-\phi)\,\text{var}(\bar{y}_{2r}) = 0$$

$$\Rightarrow \phi_{opt} = \frac{\text{var}(\bar{y}_{2r})}{\text{var}(\bar{y}_{2u}) + \text{var}(\bar{y}_{2r})}$$

Substituting ϕ_{opt} in $\text{var}(\bar{y}_{com})$:

$$\text{var}(\bar{y}_{opt}) = \left[\frac{1}{\text{var}(\bar{y}_{2u})} + \frac{1}{\text{var}(\bar{y}_{2r})}\right]^{-1}$$

10 Non Response

In all the cases that we have considered before, the assumption was that a random sample of size n , all individuals in the sample provide measurement or are observed. However in complex surveys, some of the individuals may not respond, or due to unavoidable reasons, it may not be possible to track down the individuals in the sample.

10.1 Types of Non Response

About three types can be identified.

1. **Unit non response:** This is the failure to obtain information from some of the units which could be due to transportation problem or the decision of the individual not to provide information.

2. **Item non response:** This happens when an individual answers questions in a questionnaire but decides not to respond to some of them.

3. **Non coverage non response:** This is the failure to locate or visit some units in the sample. This is a problem with area sampling in which the interviewer must find and list dwellings in a city block, i.e the target population. It arises also from the use of incomplete list of units (frame), sometimes weather or poor transportation facilities make it impossible to reach certain units during the period of survey.

10.2 Fixed Population Model

The population and the sample can be divided into two; The respondents (R) and non respondents (M), i.e., $N = R + M$.

Let:

$$\bar{Y} = \frac{1}{N}\sum_{i=1}^{N} Y_i = \frac{1}{N}\left[\sum_{i=1}^{R} Y_i + \sum_{j=R+1}^{N} Y_j\right] = \frac{1}{N}\left[R\bar{Y}_R + M\bar{Y}_M\right]$$

where:

$$\bar{Y}_R = \frac{1}{R}\sum_{i=1}^{R} Y_i \quad \text{(mean of the respondents)}$$

$$\bar{Y}_M = \frac{1}{M}\sum_{j=1}^{M} Y_j \quad \text{(mean of the non respondents)}$$

10.3 Bias of $\bar{y}_r$

Suppose a SRS of size n is obtained from the population and let r individuals respond and $m = n - r$ fail to respond, then:

$$\bar{y}_r = \frac{1}{r}\sum_{i=1}^{r} y_i$$

$$E(\bar{y}_r) = E\left(\frac{1}{r}\sum_{i=1}^{r} y_i\right) = \frac{1}{R}\sum_{i=1}^{R} Y_i = \bar{Y}_R$$

The bias of $\bar{y}_r$ as an estimator of $\bar{Y}$ is:

$$\begin{aligned}
\text{Bias}(\bar{y}_r) &= E(\bar{y}_r) - \bar{Y} = \bar{Y}_R - \bar{Y} \\
&= \bar{Y}_R - \left(\frac{R}{N}\bar{Y}_R + \frac{M}{N}\bar{Y}_M\right) \\
&= \left(1 - \frac{R}{N}\right)\bar{Y}_R - \frac{M}{N}\bar{Y}_M \\
&= \frac{M}{N}\left(\bar{Y}_R - \bar{Y}_M\right)
\end{aligned}$$

Since $N - R = M$.

We observe that the bias is small if either:

- M is small in relation to N, which implies the non response rate $\frac{M}{N}$ is small, or
- $\bar{Y}_R$ is close to $\bar{Y}_M$ (i.e., non respondents and respondents are similar in the characteristic under study).

In practice, the non response rate can be as high as 20%, 40%, or even 50%. For these cases, $\bar{Y}_R$ and $\bar{Y}_M$ are likely to be different, causing substantial bias.

10.4 Bias of $\bar{y}_r$ in a Stratified Population

Let N_h be the number of units in stratum h and $\sum N_h = N$. Let R_h be the number of the respondents in the hth stratum and M_h be the number of non respondents in that stratum such that $R_h + M_h = N_h$.

Let $\bar{Y}_h$ be the hth stratum mean,

$$\bar{Y}_{Rh} = \frac{1}{R_h}\sum_{i=1}^{R_h} Y_i \quad \text{(mean of respondents in stratum } h\text{)}$$

$$\bar{Y}_{Mh} = \frac{1}{M_h}\sum_{j=1}^{M_h} Y_j \quad \text{(mean of non respondents in stratum } h\text{)}$$

Let $W_h = \frac{N_h}{N}$ be the weight of the hth stratum.

$$\bar{Y} = \sum_{h=1}^{H} W_h \bar{Y}_h$$

$$\bar{Y}_h = \frac{1}{N_h}\left[\sum_{i=1}^{R_h} Y_{hi} + \sum_{j=1}^{M_h} Y_{hj}\right] = \frac{R_h}{N_h}\bar{Y}_{Rh} + \frac{M_h}{N_h}\bar{Y}_{Mh}$$

If we use $\bar{y}_r = \sum_{h=1}^{H} \frac{n'_h}{n}\bar{y}_{rh}$ where n'_h is the number of sample units in stratum h and $\bar{y}_{rh}$ is the mean of respondents in stratum h, the bias is:

$$\text{Bias}(\bar{y}_r) = \sum_{h=1}^{H} W_h \frac{M_h}{N_h}(\bar{Y}_{Rh} - \bar{Y}_{Mh})$$

If $\bar{Y}_{Rh} = \bar{Y}_h$ then $E(\bar{y}_r) = \bar{Y}$. To get the variance:

$$\text{var}(\bar{y}_r) = \text{var}_1[E_2(\bar{y}_r)] + E_1[\text{var}_2(\bar{y}_r)]$$

$$\text{var}_1[E_2(\bar{y}_r)] = \text{var}_1\left(\sum_{h=1}^{H} \frac{n'_h}{n}\bar{Y}_{Rh}\right)$$

$E_1[\text{var}_2(\bar{y}_r)] = E_1 \frac{1}{n^2} \sum_{h=1}^{H} \left[\text{var}_2\left(\sum_{i=1}^{r_h} y_{hi}\right) + \text{var}_2\left(m_h \bar{y}_{rh}\right)\right]$
But $\sum_{i=1}^{r_h} y_{hi}$ is constant at second phase, implying the variance is zero and:

$$\text{var}_2\left(m_h \bar{y}_{rh}\right) = \frac{m_h}{m_h} S_{rh}^2$$

because of SRSWR.

11 Imputation

Imputation is the process of replacing missing data with substituted values.

11.1 Types of Imputation

11.1.1 Deductive Imputation

In this case the missing value can be deduced with high probability. For instance, suppose the marital status of the interviewee is missing and his age is 5 years, then if we impute that the status as single, the imputation will have high correctness.

11.1.2 Mean Value Imputation

In this method, for every missing value in class h we impute the value $\bar{y}_{rh}$, the mean of all respondents in that class. The disadvantage of this method is that it can distort distribution hence inflating the variance (it reduces the estimated variance artificially).

11.1.3 Hot and Cold Deck Imputation

Here the missing value is imputed from a donor with similar characteristics. The donor is selected from the current sample (hot deck) or from a previous survey (cold deck). A common method is to sort the data by auxiliary variables and then use the previous respondent's value to impute the next missing value.

11.1.4 Nearest Neighbour Imputation

Here the missing value is imputed using the value from the unit closest to it in terms of auxiliary variables. For each missing value, find the respondent with the most similar auxiliary characteristics and use that respondent's value to impute.

Auxiliary value (x_i): $x_1, x_2, \ldots, x_i, \ldots, x_n$ Sample value (y_i): $y_1, y_2, \ldots, \text{missing}, \ldots, y_n$

Find j such that $|x_j - x_i|$ is minimized among respondents, then impute y_j for the missing y_i.

12 Variance Estimation Techniques

We have considered before estimators of the mean, total and ratio where we obtained the variance of the estimators easily. The estimators of variances of the estimators have also been established, where some of them are biased and others are unbiased. Variance estimation in survey sampling is important for constructing confidence intervals where the true variance is unknown. It also helps us to find the estimate of the variance of an estimator. Such estimators help us to find the precision of the estimator and hence be in a position to find the efficiency of the estimator.

If $\hat{\theta}$ is the estimator of θ , then the confidence interval (C.I) is given by

$$\hat{\theta} \pm 2\sqrt{\widehat{\text{var}}\left(\hat{\theta}\right)}$$

for approximate 95% C.I (since $1.96 \approx 2$).

The variance estimator $\widehat{\text{var}}\left(\hat{\theta}\right)$ should satisfy the conditions:

1. It should be unbiased or nearly so.
2. It should be stable i.e the variance of the variance estimator should be small.
3. It should be non negative.
4. It should produce confidence intervals that cover θ with probability approximately equal to the stated confidence level.

12.1 Replication Methods for Variance Estimation

Suppose $\hat{\theta}$ is an estimator of θ, based on a sample of size n. Consider T subsets of the sample where a separate estimator of θ is obtained from each subset. Let the full sample be S and the subsets be $S_1, S_2, \ldots, S_T$ such that $S = \cup_t S_t$. Let $\hat{\theta}$ be the estimate from the full sample and $\hat{\theta}_t$ be the estimate based on subset t. The alternative full scale estimator of θ can be found by averaging over $\hat{\theta}_t$:

$$\hat{\theta}^* = \frac{1}{T}\sum_{t=1}^{T}\hat{\theta}_t$$

The objective is to estimate $\text{var}(\hat{\theta})$. We consider the following two expressions:

$$\hat{V}_1 = \frac{1}{T(T-1)}\sum_{t=1}^{T}\left(\hat{\theta}_t - \hat{\theta}^*\right)^2$$

and

$$\hat{V}_2 = \frac{1}{T(T-1)}\sum_{t=1}^{T}\left(\hat{\theta}_t - \hat{\theta}\right)^2$$

We note that:

$$\hat{V}_1 = \frac{1}{T(T-1)} \sum_{t=1}^{T} \left(\hat{\theta}_t^2 + \hat{\theta}^{*2} - 2\hat{\theta}_t \hat{\theta}^* \right)$$

$$= \frac{1}{T(T-1)} \sum_{t=1}^{T} \hat{\theta}_t^2 + \frac{\hat{\theta}^{*2}}{T-1} - \frac{2\hat{\theta}^{*2}}{T-1}$$

$$= \frac{1}{T(T-1)} \sum_{t=1}^{T} \hat{\theta}_t^2 - \frac{\hat{\theta}^{*2}}{T-1}$$

$$E(\hat{V}_1) = \frac{1}{T(T-1)} \sum_{t=1}^{T} \left[\text{var}(\hat{\theta}_t) + E^2(\hat{\theta}_t) \right] - \frac{1}{(T-1)} \left[\text{var}(\hat{\theta}^*) + E^2(\hat{\theta}^*) \right]$$

But

$$\text{var}(\hat{\theta}^*) = \text{var}\left(\frac{1}{T} \sum_{t=1}^{T} \hat{\theta}_t \right) = \frac{1}{T^2} \sum_{t=1}^{T} \text{var}(\hat{\theta}_t) + \frac{1}{T^2} \sum_{t \neq t'} \text{cov}(\hat{\theta}_t, \hat{\theta}_{t'})$$

If the subsets are independent (which they usually are not in survey sampling), then the covariance terms vanish and $\text{var}(\hat{\theta}^*) = \frac{1}{T^2} \sum \text{var}(\hat{\theta}_t)$. In that case, $E(\hat{V}_1) = \frac{1}{T} \sum \text{var}(\hat{\theta}_t) - \frac{1}{T} \text{var}(\hat{\theta}^*) = \frac{T-1}{T^2} \sum \text{var}(\hat{\theta}_t)$, which is not generally equal to $\text{var}(\hat{\theta})$.

Now, note that:

$$\hat{V}_2 = \frac{1}{T(T-1)} \sum_{t=1}^{T} \left[(\hat{\theta}_t - \hat{\theta}^*) + (\hat{\theta}^* - \hat{\theta}) \right]^2$$

$$= \frac{1}{T(T-1)} \sum_{t=1}^{T} (\hat{\theta}_t - \hat{\theta}^*)^2 + \frac{1}{T-1} (\hat{\theta}^* - \hat{\theta})^2 + \frac{2}{T(T-1)} (\hat{\theta}^* - \hat{\theta}) \sum_{t=1}^{T} (\hat{\theta}_t - \hat{\theta}^*)$$

Since $\sum_{t=1}^{T} (\hat{\theta}_t - \hat{\theta}^*) = 0$, the cross term vanishes, giving:

$$\hat{V}_2 = \hat{V}_1 + \frac{1}{T-1} (\hat{\theta}^* - \hat{\theta})^2$$

12.2 Using the Variance Estimator under Sampling With Replacement to Estimate Variance under Sampling Without Replacement

Consider the variance under simple random sampling without replacement for the total:

$$\hat{Y} = N\bar{y} = \frac{N}{n} \sum y_i$$

$$\text{var}(\hat{Y}) = N^2 \frac{1-f}{n} S^2$$

$$\widehat{\text{var}}(\hat{Y}) = N^2 \frac{1-f}{n} s^2$$

Suppose the sampling fraction is small so we use the with-replacement variance estimator:

$$\hat{V}_0 = \frac{N^2 s^2}{n}$$

Then $E(\hat{V}_0) = \frac{N^2 S^2}{n}$, and the relative bias of $\hat{V}_0$ is:

$$\frac{E(\hat{V}_0) - \text{var}(\hat{Y})}{\text{var}(\hat{Y})} = \frac{\frac{N^2 S^2}{n} - \frac{(1-f)N^2}{n} S^2}{\frac{(1-f)N^2}{n} S^2} = \frac{f}{1-f}$$

For small sampling fractions, this bias is negligible.

12.2.1 Exercise

We know that under PPS without replacement sampling,

$$V = \frac{1}{2N^2} \sum_{i=1}^{N} \sum_{j \neq i} (\pi_i \pi_j - \pi_{ij}) \left(\frac{Y_i}{\pi_i} - \frac{Y_j}{\pi_j} \right)^2$$

For PPS with replacement,

$$V_0 = \frac{1}{n} \sum_{i=1}^{N} P_i \left(\frac{Y_i}{P_i} - Y \right)^2$$

and

$$\hat{V}_0 = \frac{1}{n(n-1)} \sum_{i=1}^{n} \left(\frac{y_i}{P_i} - \bar{y} \right)^2$$

If $\hat{V}_0$ is used to estimate V, show that $E(\hat{V}_0) = \frac{n}{n-1}(V_0 - V_1)$, where $\pi_i = nP_i$.

Under what conditions does the t-distribution hold exactly? The t-distribution holds exactly if the observations y_i are themselves normally distributed and the population is infinite.

How does departure from these conditions affect the distributions? Moderate departures from normality do not affect it greatly, but for small samples with very skew distributions other special methods are needed.

12.3 Example: Signatures Petition

Signatures to a petition were collected on 676 sheets each sheet had enough space for 42 signatures, but on many sheets, a smaller number of signatures had been collected. The numbers of signatures per sheet were counted on a random sample of 50 sheets (about a 7% sample). The results are given in the table below.

y_i	42	41	36	32	29	27	23	19	16	15	14	11	10
f_i	23	4	4	1	1	1	2	1	1	2	2	1	1

Estimate the total number of signatures to the petition and the 80% confidence limits.

Solution:

The sampling unit is a sheet, and the observations y_i are the numbers of signatures per sheet. Since about half the sheets had the maximum number of signatures (42), the data are presented as a frequency distribution. The original distribution appears to be far from normal, the greatest frequency being at the upper end, but there is reason to believe from experience that the means of samples of 50 are almost normally distributed.

We have:

$$n = 50, \quad \sum f_i y_i = 1471, \quad \sum f_i y_i^2 = 54497$$

$$\bar{y} = \frac{1471}{50} = 29.42$$

Hence the estimated total number of signatures is:

$$\hat{Y}_T = N\bar{y} = 676 \times 29.42 = 19888$$

For the sample variance s^2, we have:

$$s^2 = \frac{1}{n-1} \left\{ \sum f_i y_i^2 - \frac{(\sum f_i y_i)^2}{n} \right\}$$

$$= \frac{1}{49}\left\{54497 - \frac{(1471)^2}{50}\right\}$$

$$= \frac{1}{49}\{54497 - 43260.82\} = \frac{11236.18}{49} = 229.31$$

The estimated variance of $\bar{y}$ is:

$$\widehat{\text{var}}(\bar{y}) = \frac{1-f}{n}s^2 = \frac{1-\frac{50}{676}}{50}\times 229.31 = \frac{1-0.074}{50}\times 229.31 = \frac{0.926}{50}\times 229.31 = 0.01852\times 229.31 = 4.247$$

$$s.e.(\bar{y}) = \sqrt{4.247} = 2.061$$

For the total:

$$s.e.(\hat{Y}_T) = N \times s.e.(\bar{y}) = 676 \times 2.061 = 1393.24$$

The 80% confidence interval limits are (using $z_{0.10} = 1.28$):

$$\hat{Y}_T \pm z_{0.10} \times s.e.(\hat{Y}_T) = 19888 \pm 1.28 \times 1393.24 = 19888 \pm 1783.35$$

Thus: $18104.65 \leq Y_T \leq 21671.35$ for the 80% confidence limits.

12.4 Bias and its Effects

In sample survey theory, it is necessary to consider the bias of estimators for two reasons. In some of the most common problems, particularly in the estimation of ratios, estimators that are otherwise convenient and suitable are found to be biased. Even with estimators that are unbiased in probability sampling, errors of measurement and non response may produce biases in the numbers that we are able to impute from the data.

It is generally assumed that $\hat{y}$ and $\bar{y}$ are normally distributed about the corresponding population values.

$$\text{var}(\bar{y})_{SRSWR} = \frac{N-1}{N}\frac{S^2}{n}$$

In the event of SRSWR, the ith unit may appear $0, 1, 2, \ldots, n$ times in the sample. Let t_i be the number of times that the ith unit appears in the sample. Then

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{N} t_i Y_i$$

Since at each draw, the probability that the ith unit is drawn is $\frac{1}{N}$, the variate t_i is distributed as binomial with number of trials n and probability $p = \frac{1}{N}$. Hence

$$E(t_i) = \frac{n}{N}$$

$$\text{var}(t_i) = n \cdot \frac{1}{N} \cdot \left(1 - \frac{1}{N}\right) = \frac{n(N-1)}{N^2}$$

Jointly the variates t_i follow a multinomial distribution and

$$\text{cov}(t_i, t_j) = -\frac{n}{N^2} \quad (i \neq j)$$

Therefore:

$$\text{var}(\bar{y}) = \frac{1}{n^2}\left[\sum_{i=1}^{N} Y_i^2 \,\text{var}(t_i) + \sum_{i\neq j} Y_i Y_j \,\text{cov}(t_i, t_j)\right]$$

$$= \frac{1}{n^2}\left[\frac{n(N-1)}{N^2}\sum Y_i^2 - \frac{n}{N^2}\sum_{i\neq j} Y_iY_j\right]$$

$$= \frac{1}{nN^2}\left[(N-1)\sum Y_i^2 - \left(\left(\sum Y_i\right)^2 - \sum Y_i^2\right)\right]$$

$$= \frac{1}{nN^2}\left[N\sum Y_i^2 - \left(\sum Y_i\right)^2\right]$$

$$= \frac{1}{n}\left[\frac{1}{N}\sum Y_i^2 - \bar{Y}^2\right]$$

$$= \frac{N-1}{N}\cdot\frac{S^2}{n}$$

This matches the earlier result.